How to read this record

Date: 2026-10-16

Status: This directory is the official statement.

`TBdecompile/record/` is the consolidated record of the ToolBook II 8.1 work. The loose sheets under `tbworkingpapers/tbwpNNNN/` are the trail those findings were written on. They are not a second current record. When a loose sheet and this directory disagree, this directory wins. Chapter 18 names every loose sheet and the section that absorbed it.

The dated folders `20261012/`, `20261014/`, and `20261015/` stay as the notes of those days. `20261016/00-consolidated.md` is the short correction this volume expands. Read those folders for the trail. Do not read `20261015/00-how-toolbook-operates.md` section 2, or `20261015/01-theories-of-operation.md` section 1, or `20261014/02-still-open.md` opening paragraph, as the runtime create path.

Grades

Grade	Meaning
Known	Seen in a binary, a live trace, or a matching export. A later observation would have to contradict the evidence, not just a reading of a listing.
Thought	Fits the evidence and is the working model. Not yet watched end to end.
Conjecture	Plausible. Not tested.
Not true	Was a working idea. Later evidence closed it. Do not search it again.
Superseded	An older note still says this. The later note wins. The older path is listed so it is not reopened by accident.

Order

File	Subject
01-corpus.md	What the books are, and which machines can run them
02-formats.md	`.tbk`, `.sbk`, `.tbx`, `.fbk`
03-runtime.md	Modules, launch seam, NIDA system layer
04-objects.md	Live object, file open, the unnamed reader
05-graphics.md	Painters and what is stored versus computed
06-openscript.md	Bytecode, handlers, the modern subset
07-debugging.md	SoftICE, WDW, Unicorn, where the cipher lives
08-naming.md	What has been named, and what a name is worth
09-modern.md	Foolbook, Foolcourse, Foolversity, TBpipeline
10-open.md	Ranked gaps, and the notes that are closed
11-keepers.md	One-boot live addresses and decompile line numbers
12-wdw-spine.md	The book-open spine and what the three traces hit
13-specimen-pages.md	P1040, P430, P420, P1170, and the TBX draw inventory
14-openscript-bytes.md	Encoding, the back-reference fix, corpus counts
15-system-books.md	`.sbk` tiers, NIDA shell, type-name table
16-cipher.md	TBCBT segment 7 and the keyed-password schemes
17-modules.md	Module roles, including the ones that are not the reader
18-absorption.md	Every loose sheet, and where its claim went

Lookup tables that are too long to typeset live in `maps/`:

File	What it is
maps/rtm-ne-to-ghidra.md	TB80RTM NE segment to Ghidra space
maps/bas-ne-to-ghidra.md	TB80BAS NE segment to Ghidra space
maps/wdw-to-ghidra.md	One boot's WDW selector:offset against the decompile

NE segment numbers are not Ghidra `CodeN` names. Segment 1 is `1000`, segment 2 is `1008`, and the usual step is 8, with gaps. Live selectors change every boot. Offsets inside a segment do not.

What this record will not hold

Binary dumps, SoftICE console logs, and personal correspondence stay out of this repository. Names of correspondents are masked. The correspondent already used in the public notes is JohnDoe1. `who-is-who` is not part of this record.

Which older sentence to trust

If a note written on or before 2026-10-15 puts `CDBSENDCREATE` between file open and the first painted object, that sentence is superseded. The correction is chapter 4. Instructor calls ordinal 51 when it creates or rewrites an object. A lesson open does not.

If `tbworkingpapers/tbwp0001/openscript_notes.md` says the keyword table decodes to "italic" or "systime", that sentence is superseded. The bug was copying bytes `0xC0–0xFF` from the output stream. Chapter 14 is the encoding that actually decodes the handlers.

Absence from a Ghidra HTML export is not absence from the binary. The listings have undisassembled runs. A candidate is closed only when the disassembly or a live trace shows what the function actually does.

The printable form of this directory is ToolBook-record-20261016.pdf.

1. The corpus

Claims

Id	Grade	Statement	Source
C-001	Known	The lesson files in this work are Nida Corporation electronics lessons, written as Click2learn ToolBook II 8.1 shells. The book version word is `01 00 00 08`. This Nida is not the drug-abuse institute that shares the letters.	TBWorldview WV-001
C-002	Known	The catalog is a Navy ATT menu export (`nida:m3search`): menu `0_MASTER3` plus 20 rating tracks. 611 `.tbk` files are on disk. 290 catalog ids are unique. 321 files are not in the catalog.	courses listing, project notes 2026-10-08
C-003	Known	The worked specimen is Logic Families, `9e954d01.tbk`. It is not self-contained. It names the NIDA system books and jumps into `nida81.tbk`.	lesson header at file `0xB1A96`
C-004	Known	Windows 98 can step the engine. 32-bit Windows XP can run it. 64-bit Windows cannot run the 16-bit NE images. Windows 11 is outside the range.	WV-028
C-005	Known	The engine calls 16-bit `USER`, `GDI`, and `KERNEL`. A startup log that reports `USER32`, `GDI32`, and `KERNEL32` missing is not a broken install.	WV-027
C-006	Not true	ToolBook 8.1, as held here, is a 32-bit product that needs Windows 10 or earlier. The images in hand are 16-bit NE. ToolBook 9.0 is the 32-bit product.	`TBdecompile/20261014/03-tbx.md`

What the set is

The books are computer-aided instruction for Navy electronics ratings. The menu export lists tracks such as GSE, EM, IC, GM, FC, ET, STG, AE, ATI, AO, ATO, AS, TM, CTM, FT, ETSS, STS, CTT, MN, and MV, at versions V03.2 through V03.4. `courses.xml` is that menu. It is not a drug-abuse curriculum.

A lesson is a ToolBook II 8.1 book plus a shell. The shell is `nida81.tbk`. The system books are `nc81base.sbk`, `nc81recs.sbk`, `nc81test.sbk`, `nc81trnr.sbk`, and `NC81BANK.SBK`, with `nc81int.dll` and `nc81rec.dll` beside them. Removing the system books leaves a blue screen, breaks resize, and stops the menu. The lesson file itself still opens. Renaming `nida81.tbk` removes the splash and reports that the shell is not a book. The lesson can still be entered.

`9e954d01.tbk` is the book every later measurement is checked against. It is Logic Families. At file offset `0xB1A96` it names `tb80r.sbk`, `nc81trnr.sbk`, `nc81base.sbk`, `nc81recs.sbk`, and `nc81test.sbk`, and it calls `initializebase`, `initializerecs`, `initializetest`, `openSession`, and `gotoPageList`. It also jumps to `nida81.tbk` and to `93954302.tbk`.

Who used it

Known. The published users of this particular set are the Navy lessons on disk and the training shops that shipped ToolBook courses. This is not a living authoring market. Someone outside the set who still holds a native 8.x book has the same file problem. That last sentence is a thought, not a census (WV-030).

Machines

The floor for a real run is Windows 98. That is the known-good debug host, not the top of the range. The ceiling is 32-bit Windows XP. The 16-bit NE images do not run on 64-bit Windows. A correspondent described 8.1 as 32-bit and 9.0 as the release that added 64-bit support. The binaries in this archive are 16-bit NE. Until a 32-bit 8.1 engine is actually in hand, the binaries win (C-006).

SoftICE work is closed as of 11 October 2026. Later tracing used the Watcom debugger on a fresh Windows 98 image. Those two debuggers do not share commands. See chapter 7.

2. File formats

Claims

Id	Grade	Statement	Source
F-001	Known	A lesson begins with a `0xC00` PE stub that has no useful code. The book stream is marked by `ZZZTBKBOOKLOC` and magic `0x22222222`, then `03 4A 42 4F` and version `01 00 00 08`.	WV-002
F-002	Known	Page records are the only objects that start `80 65 00 05 00 02 02`. Widgets inside a page start `14 00 00`.	WV-003
F-003	Known	`.tbk` and `.sbk` are the same container. The page name sits at offset `0x0B` of the page record.	WV-031
F-004	Known	Field text is a `Text\0\0` property: 8 bytes, a 16-bit length, then the string. The string often starts with CRLF.	page reader notes
F-005	Known	A ToolBook 11.5 Save To XML, enabled by `saveSpecial=true`, writes a folder of XML plus resources. A `.tbkx` file is a ZIP of that folder.	`TBdecompile/20261014/03-tbx.md`
F-006	Known	Binary 8.1 stores many draw objects as bounds. The 11.5 XML expands those bounds into Vertices lists.	P1040 pin bounds `5055,4605,5205,4860` match the TBX Vertices start
F-007	Known	Foolbook `.fbk` / `.fsk` is a 4-byte magic (`FBK1` / `FSK1`), one flags byte, then JSON. Bit 0 means the payload is zlib-compressed.	Foolbook format notes

2.1 Native book

`tbload.exe`, through `MINSTB32.dll`, searches for `ZZZTBKBOOKLOC`. The signature check is a virtual call, not a literal compare sitting in `tbload`. On `9e954d01.tbk` the JBO body is at file offset `0xC00`. A later open measurement put the same book's first useful read at `0x9EC`. Both figures are about the same file. `0xC00` is the stub length. `0x9EC` is where one host's stage-2 open landed. Do not treat them as two different formats.

A page record holds a 32-byte name, usually `P` plus digits, then a property block. The property block at +43 of the early notes starts `14 00 00 01`. Widgets are not separate top-level objects on disk. They are length-prefixed chunks nested in the page.

Known. An ascending table of words inside a page, such as the 27 words at page `+0x8C3` on P1170, is an offset directory into a `14 00 00 83` content blob. It is not a Group. A Group is its own object and has its own member list.

Not true (WV-010). On-disk `14 00 00 01` does not become `14 00 00 11` when the book loads. One early dump was read that way. Later live objects used `14 00 00 02`. `9e954d01.tbk` has both `01` and `11` on disk. `93950101.tbk` is the other way around. That is a file difference, not a load-time mutation.

Property ids recovered from `TB80BAS` resource type `0x2D1`, id `0x2D0`, are `0x4000` plus a slot:

Id	Name
`0x4004`	bounds
`0x4015`	idNumber
`0x4023`	script
`0x402E`	text
`0x4034`	vertices

`0x4050` is the runtime property id of the point-array setter. It is not a record marker on disk.

2.2 System books

`.sbk` files use the same container. The NIDA `.sbk` files are resource containers: the object-type name table (ComboBox, Stage, Picture, Hotword, RecordField, PaintObject, Curve, AngledLine, Pie, Arc, IrregularPolygon, Polygon, Line, RoundedRectangle, Ellipse, Group, Field, Button, Rectangle, then Page, Background, Book) and menu resources. They are password-protected. The password is not in this record, and the decompile does not need it.

Thought, narrowed. A search for the plain words `to handle` finds no OpenScript in those `.sbk` files. A later reader found compressed handler streams in `tb80r.sbk` (33 handlers), stored as keyword opcodes rather than the words `to handle`. “No script at all” is too strong. What those streams do when a lesson runs has not been traced. `nc81base`, `nc81recs`, `nc81test`, and `nc81trnr` still have no decoded `to handle` text.

`nida81.tbk` is a real book, not a resource bag. It has 43 objects and pages `P20000` through `P22130`. It does not name lesson ids. It has 100 `buttonClick` handlers in the same bytecode as a lesson. P310 is Minimum Requirements (586, 133 MHz, 12 MB RAM, 1 GB, mouse, Windows 98). P20001 is the small menu. P22130 is Test Console Setup.

2.3 The 11.5 export

Known, as of the exported specimen. In `Instructor115.ini`, under `[Options]`, set `saveSpecial=true`. Open a copy of the 8.1 book, answer the version warning, press F3 for Author level, then File → Save To XML. The result is a folder named `.TBK TBX`. A `.tbkx` is a ZIP of that folder. Opening the ZIP in 11.5 imports it and rebuilds a book.

The schema says Bounds is required on a draw object and Vertices is optional. The schema spells IrregularPolygon wrong (`IrregularPloygon`). The binary stores bounds. The export expands them to Vertices. On P1040 the pin bounds `5055, 4605, 5205, 4860` match the start of the TBX Vertices list. That is the expansion direction: bounds in the 8.1 file, point list in the XML.

The exported specimen of `9e954d01` has 131 `Page.xml` files, 14 backgrounds, about 75 BMPs, and 302 `.script` files. Two of those pages, P440 and P2920, have no `80 65` page record in the 8.1 binary. The binary has 129 pages. All 129 names match a TBX page. The two extras are an export addition, not a missed record.

Saving from the 11.5 trial converts the book. An 8.1 runtime cannot open that converted file. 8.1 stores an imported picture as a basic blob and turns it into a BMP. 11.5 stores WMF, JPG, TIF, BMP, PNG, and GIF as themselves.

WV-009 called the export a conjecture on 12 October. The specimen exists. The conjecture is closed. WV-013’s description of the ZIP-plus-XML shape matches the specimen.

2.4 Foolbook envelope

`.fbk` and `.fsk` are not renamed JSON files with a new extension. The envelope is magic, one flags byte, then a JSON payload. The engine reads compressed, uncompressed, and the older plain-JSON working copies. `.json` remains the working copy. `.tbk` and `.sbk` stay supported so an existing ToolBook file can be opened. System books load core first, then user `.fsk` files as mods.

The philosophy written for that reader is discipline with a recorded deviance: use explicit points when they exist, then the TBX Vertices, then a bounds approximation, and say which one was used.

3. Runtime

Claims

Id	Grade	Statement	Source
R-001	Known	The intended launch is `TB80RUN` (32-bit) spawning `TB80NET.EXE` with the book on the command line. The seam is that spawn, not an in-process thunk.	module catalog, `TB80RUN` survey
R-002	Known	`TB80RTM.EXE` is the 16-bit shell. It owns `TBKMain`, `TBKClient`, `TBKChildWindow`, `TBKReserved_1`, `SchedulerWndTB80`, and the frame class created with `TB80MAIN`. A lesson click is delivered to `TBKClient`.	SoftICE class list
R-003	Known	`TB80MAIN.EXE` is a 23552-byte 32-bit stub. It calls `TB80RMT CreateRemoteCallServer` and pumps keys. It is not the book engine. Failure text: “TB80MAIN: Loading ActiveX support failed.”	`TB80MAIN` decompile
R-004	Known	`TB80BAS` is the book server (“ToolBook Server DLL”), 606 exports. `CDBINIT` is ordinal 1. `CDBFILEOPEN` is ordinal 8. `CDBVIEWPAGE` is 15. `CDBVIEWBKGND` is 18. `CDBPAINTCLIENT` is ordinal 26.	export table, WV-011
R-005	Known	`TB80RTM` `WM_PAINT` calls `CLIENTTOPAGE`, then ordinal 26. Shape drawing is in BAS, not in RTM.	`tbwp0018/rtm-paint-path.md`
R-006	Known	`TB80RMT` is not on the path that opens a lesson. The 1.5 `PHYSSIZE` index is not in these 8.1 books.	WV-014
R-007	Known	`nc81int.dll` exports 47 trainer calls. `IntCheckComm130E` opens a port at 9600 8O1, sends a probe, and returns `0x3D` on match or `0x3C` on failure. Families: 130E, TD2000, 5050, 205E.	`nc81int` exports
R-008	Known	`nc81rec.dll` writes `0x46`-byte session records to `SESSION3.REC`. Exports include `RecStartSession`, `RecCloseSession`, `RecBookmarkSession`, `RecWriteResponse`, `RecReadVariable`.	`nc81rec` exports

3.1 Process

`tbload.exe` finds the book marker and hands off through a vtable at `FUN_00406a70` (virtual call at `00406aa1`, vtable+4). That vtable is not resolved. **Thought:** that call is the first handoff out of the 32-bit search.

`TB80RUN` does not open the book itself. It starts `TB80NET.EXE`. `TB80NET` is a 16-bit NE executable, not a DLL. It is the network access provider (zlib, `TBKNET3WNDPROC`, `TBK.INI`). It does not parse pages.

`TB80W32.DLL` (about 246 exports) and `TB80W16.DLL` (about 406) are the thunk halves. W16 is also the `DoCmd*` command bus (`DoCmdDraw`, `DoCmdMouseDown`). `TBFILE32.DLL` supplies long filenames to the 16-bit engine. `TB80F32L.dll` is the flat 32-bit entry and the common-dialog hook.

`TB80RTM` is the window. Classes seen live include `SchedulerWndTB80` and `TbkNet3TB80` (the latter owned by `TB80NET`). The lesson click does not arrive on `TBKChildWindow`. Break the `TBKClient` window procedure. `BMSG` does not see that 16-bit queue.

3.2 Book server and neighbors

Module	What it is for	What it is not
`TB80BAS.DLL`	Objects, properties, painters, twip scale, file open	Not the window
`TB80CMP.DLL`	OpenScript compile and expand. Resources `0x77` and `0x78`	Not the page walker
`TB80UTL.DLL`	Type-name table, string manager, metrics, heaps	Not the `14 00 00` reader
`TB80RCR.DLL`	Linked beside the book server. Exports only partly named	Not classified
`TB80ENU1.DLL` / `ENU2`	English strings, 16-bit and 32-bit	Not code that opens a book
`TB80BMP.DLL`	DIB, metafile, FX blit. Ordinal 3 is the bitmap reader	Not a picture store
`TB80JPG.DLL`	JPEG. Version resource still says `TB60JPG.DLL`	
`TB80LNL.DLL`	MCI, wave, MMIO. The notify callback is Lionel	
`TB80MM.DLL`	`tbkmmNotify`, `imageNotify`, `mciNotify`, `timerNotify`	Does not classify objects
`TB80OLE` / `OCX` / `ACX` / `VBX`	Clipboard, controls, automation	Not on the Logic Families paint path

`linkDLL` libraries add commands only when a handler links them: `TBDLG`, `TBWIN`, `TBDOS`, `TBREG`, `TBDB3`, `TBPDX`, `TBCBT`. `TBCBT` is the CBT scoring extension. Its segment 7 holds the password cipher. See chapter 7. `TBWIN` is `INIT_TBKWIN` and `TBM_EVALUATE`.

Neither TBWIN nor TB80NET walks page records.

Author binaries TB80EDT.DLL and TB80.EXE are not in this set. The runtime half of the author contract is visible: BAS exports CDBEDITBOOK, CDBVIEWBOOK, CDBISVIEWERINGENERICEDITMODE, CDBSENDEDITLEAVETOALL*, and TBXAUTHORMODE. RTM calls GETAUTHORLOCALIZEDDLL. That is bookmarked, not solved.

3.3 NIDA layer

A system book adds script above the book. A linkDLL adds commands. Most of what an author treats as “ToolBook” is one of those two, not a new entry point in BAS.

tb80r.sbk is the Asymetrix reader book (Help, Bookmarking, Logging, Path manager, Popup Page Manager). It is not the engine. tb80hyp.sbk is hyperlink and help. tb80anm.sbk is path animation.

The lesson calls initializebase, initializerecs, and initializetest by name. **Thought.** Those names have not been watched as a call log, and the order in toolbook.lst has not been watched live. version.ini in the set is 03.001.0006, dated 04-18-05. CAILT.DAT maps course ids to Navy CDs. 9E954 maps to DIE. STDAPPS.LST is a short encoded list, not a book.

nc81int.dll and nc81rec.dll are 32-bit PE. They are not Asymetrix. The trainer probe and the session file are the two behaviors that matter for getting a NIDA lesson to run as a lesson rather than as a picture of a page.

3.4 What startup has been watched

WDW on W16, before any book open, hits platform detection, OLE init, and the 32-bit thunk: GETPLATFORMREF, IS32BITENV, GETRUNTIMELOCALIZEDDLL, ACXINITIALIZETOOLBOOKSUPPORT, ACXOLEINITIALIZE, LOADLIBRARYEX32W, GETPROCADDRESS32W.

WDW on UTL, after INITTASK, hits string and task utilities: GETSTRMGRSTRING, plugin init, TBK.INI profile reads, metrics, GHEAPCREATE. No record walk.

The Unicorn host has not opened a book. Pass 06 asks for TB80.INI and exits. See chapter 7.

4. Objects, file open, and the reader

Claims

Id	Grade	Statement	Source
O-001	Known	A live object is `0x76` bytes. Bounds are four words at `+0x0F`, in twips. At 96 dpi, screen pixels are stored / 15, which is `(stored * 96 + 720) / 1440`.	live traces, `FUN_1008_01dd`
O-002	Known	`+0x1F` selects the BitBlt raster op. `+0x3B` is a GDI handle. `+0x30` is the point count. `+0x32` is the point-array far pointer, 4 bytes per point, stored twips.	polygon and BitBlt traces
O-003	Known	`CDBSENDCREATE` (BAS ordinal 51, `12c8:00BC`) creates or rewrites objects from the Instructor UI. The runtime book-open path does not call it. `FUN_12c8_0ef8` is downstream of that author path.	`tbwp0025/cdbsendcreate-is-author.md`, WV-042
O-004	Superseded	The chain `CDBSENDCREATE` → `FUN_12c8_0850` → `FUN_12c8_0e32` → `FUN_12c8_0ef8` is the runtime reader.	`TBdecompile/20261015/00-how-toolbook-operates.md` §2, `tbwp0002`, `tbwp0008`
O-005	Known	SoftICE and WDW both reached `OBJFILEOPEN` and the buffered read in `FUN_1160_03de`. The raw DOS path at offset `04F3` is skipped when the buffer is already satisfied.	2026-10-16 consolidated note
O-006	Known	The W16, UTL, and BAS traces in `tbwp0026` do not contain a function that reads a `14 00 00` record. `CDBCHUNK*` walks text. `VALUENEWLONG` constructs a value. `OBJFILEOPEN` is downstream of the reader.	`tbwp0026/trace-review.md`, WV-044
O-007	Thought	The runtime reader walks nested chunks by length, classifies them, and builds objects without calling `CDBSENDCREATE`. It sits in BAS after `CDBFILEOPEN` and before the first paint. It has no name.	open-question notes

4.1 The live object

These offsets are the ones the papers and the live traces agree on.

Offset	What	Grade
`-2`	Type byte. `0x11` is an arc. The other type bytes are not mapped.	Known for `0x11` only
`+0x0F`	Bounds: left, top, right, bottom. Written as twips.	Known
`+0x17` `+0x19` `+0x1B` `+0x1D`	Four words copied by `FUN_12c8_0ef8` from a source structure at `+0x3E` `+0x40` `+0x42` `+0x44`. The disk `14 00 00` record does not sit at those source offsets.	Known as a copy. The source is not the disk record.
`+0x1F`	BitBlt raster-op flag	Known
`+0x30`	Point count (word)	Known
`+0x32`	Point array far pointer	Known
`+0x3B`	GDI handle	Known
`+0x52`	Per-pin index. Not applied by the current renderer. Pins that share point bytes draw on top of each other.	Known that the field exists. Not known how paint uses it.

`ScaleTwipsToScreen` is `FUN_1008_01dd`, BAS segment 2 offset `0x1DD`. With `DS:01A6` and `DS:01A8` set to 96 it turns stored twips into screen points. On the P1140 to P1130 change those words were 96 and 96, so the short form is division by 15.

`FUN_12c8_0ef8` copies four words into the object. It does not parse a page. The disk record and the structure it copies from are different shapes. The step between those two shapes is still unnamed. That step is not `CDBSENDCREATE`.

4.2 Author create, stated once

`CDBSENDCREATE` receives bounds that are already formed. It does not compare a buffer to `14 00 00`. Type dispatch inside it selects among new-object and bounds-checked paths. The property registrar `FUN_1058_0a98` writes property id `0x4034`. It does not write painter pointers.

Nothing inside BAS calls `CDBSENDCREATE` by name. One external thunk from `FUN_13d0_0822` reaches ordinal 51. That caller is author UI state (it tests bytes such as `0x26`, `0x36`, `0x38` and stores coordinates around object `+0x638`). It is not the book-open loop.

Do not break on ordinal 51 to find the reader. The stop will fire when Instructor creates or rewrites an object. It will not fire because a lesson page was opened.

Older notes that drew this chain as the runtime path:

- `tbworkingpapers/tbwp0002/bas-create-path.md`
- `tbworkingpapers/tbwp0008/cdbsendcreate-sites.md`
- `TBdecompile/20261015/00-how-toolbook-operates.md`, section 2
- `TBWorldview/ToolBook_Architecture.md`, the book-open flowchart that calls `CDBSENDCREATE` per widget

Those files still describe a real function. They mis-place it.

4.3 File open, as far as it has been stepped

`CDBFILEOPEN` is BAS ordinal 8. **Thought (WV-007).** It is the export on the open path. It has not been shown to expand a `14 00 00` record itself.

Under it, `OBJFILEOPEN` has a large frame (`0x25A` in one trace). `FUN_1160_03de` is the read wrapper. It calls `FUN_1258_00a8` (buffered path). If that call satisfies the request, a jump skips the raw DOS read at offset `04F3`. Live dumps of a 4-byte read returned an internal pointer (`FC B5 BD B0` at one buffer), not a page header and not the JBO signature. A wider dump showed `4F 46` nearby. That is not a page record.

`CDBCHUNKFIND`, `CDBCHUNKLAST`, `CDBCHUNKEND`, and `CDBSKIPSEP` walk characters: comma, CR, LF, tab, space. They are text chunkers. A decompile that shows `ISDBCSLEADBYTE` and a comma test is not the page walker.

4.4 Where the reader is not

Closed, so they are not searched again as the reader:

Candidate	What it actually is
`FUN_1028_2680`	Filter registrar. Calls `TBREGISTERFILTER`. The `0x14` is a type-code limit, not the record marker.
`FUN_1008_2d31` (switch at `1008:2DC7`)	OpenScript math. `+ - * /`, `abs`, `sqrt`, `log`, `exp`, `sin`, `cos`, `atan`, `asin`, `acos`.
`TB80RTM`	No named file open. Paint goes to BAS ordinal 26.
BAS segments `1018`, `1278`, `1478`, `14b8`, `1508`, `10d0`, `1030`, `14a8`, `1160`, `1070`	Sampled. No stream walker. `1160` is debugging (`CDBGETNEURONURL`, `CDBATTACHBREAKADDRESS`).
`FUN_10b0_00a6`	Dispatches on type codes including `0x14`. It is a type dispatcher, not a reader.
Code at `12e8:0000`	OLE object clone.
BAS segment 182	Real preload code, 39 far calls with a zero segment and no reloc. Not on the book-open path.

Known. No static search of the disassembled image has found the bytes `14 00 00` used as a compare against a far pointer that then builds an object. Thirteen undisassembled runs contain the word `14 00`. Those are far-call immediates, `MOV CX, 0x14`, or jump displacements.

Thought. The reader trusts the length-prefixed layout and does not search for the marker. That would explain the missing literal. It is not proven. The way to prove it is a break after `CDBFILEOPEN`, before the first paint, on a lesson page, with the page blob address in hand.

4.5 Page and background

Known. `80 65` records are the pages. `9e954d01.tbk` has 129 of them.

Known. A TBX page can name a background by an id, such as 23 on P10213, that does not appear as a word or a dword in the matching binary page. The link is indirect. Parent and z-order are not in the page record that has been decoded. Absolute bounds are not stored for every shape. The P1170 bracket was computed from bounds relative to a parent. The parent offset has not been applied to the P420 rects.

5. Graphics

Claims

Id	Grade	Statement	Source
G-001	Known	Polygon kind `04 00`: fill dword at `+4`, `+0x0C` = `04 00`, count word at `+0x1A`, twip pairs at `+0x22`. Screen = twips / 15.	P430 render, live `GDI!POLYGON`
G-002	Known	The polygon painter is BAS `FUN_1408_0376` (segment `0x82` offset `0x376`). It copies object `+0x30` / `+0x32` to `DS:30AC` and calls `GDI!POLYGON`.	SoftICE, 2026-10-10
G-003	Known	Content rectangles are bounds-only. In the page blob the tail is four twip words, a zero dword, a fill dword, then `01 00 00 04`. `9e954d01.tbk` has 961 of that marker.	Foolbook `shapes.py` count, P1040
G-004	Known	An arc object is type `0x11`. Four twip pairs at `+0x25`, `+0x29`, `+0x2D`, `+0x31`. Bounds at `+0x0F`. Fill at `+0x18`. Painter is BAS segment `0x7B`.	live arc, 2026-10-10
G-005	Known	ToolBook does not call `GDI!PIE`. Pie wedges are computed by `GDIELLIPSEPOINTFROMANGLE` (property `0x4123`) and filled with the arc geometry.	import table and decompile
G-006	Known	A diagram bracket on P1170 is not stored as points. BAS segment `0x5B` computes four polyline points from the object bounds.	live `GDI!POLYLINE`, 4 points
G-007	Thought	P420's flame is several pen strokes in the record at page `+0xA13`, matching three 96-point red IrregularPolygons in the TBX export. The on-disk control structure is not known.	render attempts, TBX

5.1 Who paints

The painters live in `TB80BAS`. They are reached through a type dispatcher, not by a named call from outside BAS. `TB80RTM` handles `WM_PAINT` only as far as `CLIENTTOPAGE` and `CDBPAINTCLIENT` (ordinal 26).

Painter	Where	GDI
`PaintPolygon`	`1408:0376`	`POLYGON`
`PaintPolyline`	`13e8:00E6`	`POLYLINE`
`PaintRectangle`	`1020:03E4`, and segment `0x83` (`FUN_1410_0066`)	`RECTANGLE`
`PaintArc` / alt	`10A0:0558`, `10A0:0458`	`ARC`
`PaintEllipse`	`1230:2F1C` and an alt	ellipse
`PaintRoundRect`	`1230:2278` and alts	round rect
`ScaleTwipsToScreen`	`1008:01DD`	none. Converts in place.

The polygon type table is not in the static image. Unpacked BAS still carries Optloader segment-table debris. The HTML listing has zero cross-references to `FUN_1408_0376`. **Thought.** The table is built at load, or fixed up by an NE relocation that the listing does not show as a pointer. A break on the painter, then one frame up, is how the table gets found. Searching the file for the bytes `76 03 08 14` will not find it. The selector is assigned at load.

`SetPointArray` (`FUN_1408_00d8`, property `0x4050`) stores the count at `+0x30`, allocates `count * 4` with `OBJNEWNEAR` into `+0x32`, and copies the points. That is the runtime setter. The file does not contain a `0x4050` record.

5.2 What is stored

Polygons. `9e954d01.tbk` has 96 kind-`0400` records on 13 pages. P430 has 22. A broader search that ignores the kind word produces false positives. P430 draws as a chip: silver pins (`0xC0C0C0`) and a light-gray body (`0xE0E0E0`). One live 11-point polygon is a pin tip, not the body. The captured screen points were `(218,335)` through `(218,341)`, a 10 by 17 notch. The same points exist in the file as twips near page offset `0x9CB` of P430: `(3147,3862)` through `(3147,3973)`.

Pins that share identical point bytes stack on top of each other until the index at `+0x52` is applied. That index is not applied in the current clean-room draw.

Rectangles. P1040 is the TTL intro page (TBX page id 880). Two DIP packages: 7400 on the right, 5400 on the left. The 8.1 file stores 28 pin boxes (fill `0xC0C0C0`) and two dark bodies (fill `0x404040` in the TBX, `64,64,64`). The pin tip is not an 11-point list on this page. Paint expands the tip. The TBX export writes the expanded Vertices. Matching a TBX Vertices list

back to the binary means matching the bounding box, not searching for 11 points.

Brackets and strokes. On P1170 the bracket's live polyline was `(354,201) (321,201) (321,238) (358,238)`. The object bounds in twips were left `4801`, top `3015`, right `5370`, bottom `3570`. Divided by 15 that is the same rectangle. The points are computed. Connecting lines and detail strokes of this kind are not extracted yet.

Arcs. Live geometry is known (G-004). Fill `0x808080` was seen at `+0x18`. No page record has been matched to type `0x11`, and the start and sweep angles are not located in the struct. Do not set a breakpoint on `GDI!PIE`. The call is not imported.

Curves. Curve's type id in `nc81base.sbk` is `0x05C7` (file offset `0x1810`). That id is not in the P420 page records. The point runs sit in the record at page `+0xA13`. A naive walk draws a tangle in screen space around 400–500. The TBX export of the flame is three 96-point red IrregularPolygons. **Thought.** The tangle is strokes concatenated that should be drawn separately. A length prefix or a zero terminator between groups has not been proven. There is no `Curve` name on the page.

Pictures. Windows BMPs can be carved. One match against the TBX export paired 67 markers with 72 bitmaps. The widget still does not name the bitmap. Compressed pictures go through `TB80BMP` ordinal 3 and have been followed as far as a far call at `0x1FCD`. `Lz77Expand` is `FUN_1000_0031` in BMP.

The shared near-start graphic in the lesson books is a 581 by 157, 8 bpp gold plate, not a text banner. Two notes locate the `nida81` splash building: file `0x413FAB` (323 by 344, 8 bpp, off-bits 310) and a carved copy taken from file `0x487BE` (27874 bytes). This record does not treat those two offsets as the same byte. The live `BITMAP` header that was found was width `0x143`, height `0x158`, 8 bpp, stride `0x258`, `bmBits` null. **Known** that the splash is a DIB. **Not known** which property on the page object holds its id.

BAS segment `1150` builds the book palette (`CREATEPALETTE`, `SELECTPALETTE`, `REALIZEPALETTE`). An 8 bpp BMP needs that palette. Segment `1018` creates and deletes pens, brushes, and bitmaps. Who owns the lifetime is open. Fill color is known. Stroke color and pen width are not.

Field text is `DRAWTEXT` from `param_5+0x4A`, `+0x4C`, and `+0x50`. The rest of the field layout is open.

5.3 What a render is allowed to claim

A clean-room picture of a page is not the engine (WV-005). The current honest draw is:

1. Explicit point records (kind `0400`) where the count and the pairs are in the file. 2. TBX Vertices where an export exists and the binary only stored bounds. 3. A bounds rectangle where neither points nor an export exist.

Say which of the three was used. P430 from kind `0400` is a check against the chip. P1040 from bounds is a check against the two packages, with the understanding that the pin tips are fatter on screen than the stored boxes. P420 from a raw point dump is not a flame until the stroke breaks are known.

6. OpenScript

Claims

Id	Grade	Statement	Source
S-001	Known	Handlers are compressed bytecode. The keyword table is NE resource `0x77` (type 100, id 119). The offset table is at byte 2 plus count times 4. There are 994 keywords.	decoder notes
S-002	Known	Bytes `0xC0`–`0xFF` copy from the compressed input stream, not from the output. An earlier decoder that copied from the output was wrong.	WV-032, decoder fix 2026-10-10
S-003	Known	The decoder used for NIDA navigation stops at the first standalone `end`, so trailing decode garbage is dropped.	engine notes
S-004	Known	`9e954d01.tbk` yields 76 handlers on 39 pages. `nida81.tbk` yields 100 `buttonClick` handlers. `tb80r.sbk` yields 33, with the `to handle` prefix also compressed.	corpus counts
S-005	Known	`FUN_1008_2d31` is the expression evaluator. The modern engine implements `+ - * /` and `abs`, `sqrt`, `log`, `exp`, `sin`, `cos`, `atan`, `asin`, `acos`, and otherwise leaves the original text.	`openscript_analysis` 1.0.0r6
S-006	Known	A `buttonClick` on `nida81.tbk` page `P20000` goes to `P20001` because the handler says so. Navigation is not “only a quoted page id.”	WV-033, WV-041
S-007	Thought	A short script compiled by Instructor 8 and expanded with the existing decoder would lock the keyword map. That compile has not been done.	`tbwp0014/instructor8-compile-test.md`

6.1 What the bytecode is

Resource `0x77` is a compiler emit routine as well as the keyword pool. Resource `0x78` (id 120) is another emit routine. `COMPILERPROPINFOFROMINDEX` locks resource `0x70`. The property-name table is a different resource (BAS type `0x2D1`, id `0x2D0`). Do not treat every locked resource in that range as the keyword list.

The byte classes, the back-reference formula, and the corpus counts are chapter 14. An early reading of the same bytes produced "italic" and "systime" and concluded the keyword table was wrong. That reading copied `0xC0`–`0xFF` from the output stream and used the wrong offset-table stride. It is superseded. Do not cite it.

High-bit opcodes (`0xE0`, `0xF0`, `0xFF`, and neighbors) emit whitespace, CR/LF, or a repeated span. After the input-stream back-reference fix, the 76 handlers in Logic Families decode. Examples that are real source, not Latin-1 of the bytecode:

- `send next`
- `go to page "P400"`
- `go to page "P310" of book "nida81.tbk"`

The common subset the courses actually use is `go to`, `send`, assignment, `put into`, `set to`, `get`, increment and decrement, nested `if` / `else` / `end if`, `step from` / `to`, and `name of this book` / `this page`. The engine stops at the first standalone `end`.

`tb80r.sbk` uses the same bytecode. The difference is that the handler name prefix is compressed too. Encoding rules match. The byte histogram has more `0xFF` and `0xE0`.

6.2 What a lesson does with it

Logic Families is a menu of jumps, not a self-running program.

Page	What the handler does
P900	Lesson menu. TTL → P1001. Other → P2001. Final quiz → P10001. Exit → `nida81` P20001. No Next.
P1001, P2001	Back returns to P900, not to `nida81` P20001.
P300	`selection1` is the IBM PC Compatible Computer line. `buttonClick` sends `lastbook nida81.tbk` and `nextpage` P310.
P10001	Assessment. Directions → `nida81` P10020. Exit → `nida81` P20001. Begin → P10210.
P10410	Section end. Menu returns to P900.
P1290	Quiz. Correct choice 5408. Again → P1292. Next → P1300.

A retry page returns to the quiz whose `again` field names it. `initializeRecs` on the Maybe2 shell clears responses when a session loads. That behavior is inferred from the shell script, not from a live `nc81rec` trace.

The modern engine also accepts `send to`, a `get` that calls, `forward`, `break`, and `property of`. Predicates such as `is not in` are recorded rather than evaluated. Property writes are not applied to a live object. The TBX `.script` files are plain text and are easier to read than the bytecode. They are a playback path for the export, not a reader of the `.tbk`.

6.3 Expressions

The 148-case switch at `1008:2DC7` is `FUN_1008_2d31`. It stays in segment `1008`. It checks for names such as `log`, then uses the FPU (`FLD`, `FSTP`, `FCLEX`). It is not the record reader and not a paint path. The clean-room evaluator copies that set of operations and falls back to the original string when it does not recognize the right-hand side.

6.4 What is not done

The full keyword map is not locked. Resource `0x77` has 994 entries. The engine proves the ones the NIDA handlers use. A compile of a short script in Instructor 8, saved into a book, then expanded, would show whether the offset table and the names still match. That test is written down and not run.

Nested property expansion, the opcodes past the common subset, and `forward` into a system-book handler are unimplemented or only sketched. `COMPILERCOMPILETEXT` is CMP ordinal 1. `COMPILEREXPANDTEXT` is ordinal 3. `CFUNEXECUTE` is 8. `CFUNEVALUATE` is 9. Those ordinals are the stable breakpoints if the compile test is run.

7. Debugging and hosts

Claims

Id	Grade	Statement	Source
D-001	Known	SoftICE and the Watcom debugger do not share command syntax. SoftICE field work closed 11 October 2026. Later breaks are WDW.	field manual, `tbwp0026/wdw-break-syntax.md`
D-002	Known	WDW has no full automatic call trace. `break resume on` plus `log` logs and continues. `resume on` is a property of a breakpoint that already exists.	`tbwp0026/wdw-trace.md`
D-003	Known	In WDW, module qualification uses `@` (`TB80BAS@OBJFILEOPEN`). SoftICE's `!` does not work there.	same note
D-004	Known	Symbol maps are `NE-segment:offset name`. The segment index is not a live selector. SoftICE and WDW do not load the `.map` as a module.	`tbwp0003/TB80BAS.map` (3799 entries), `TB80RTM.map` (2669)
D-005	Known	The Unicorn host has not opened a lesson. Pass 06 exits after `TB80.INI`. The entry does not ask for the book.	WV-006, WV-036
D-006	Known	TBCBT NE segment 7 holds `KEYEDPASSWORDENCRYPTEX`, `MAKEKEYEDPASSWORDEX`, `ENCRYPTEX`, and `DECRYPTEX`.	WV-043
D-007	Known	NE segment numbers are not Ghidra's `CodeN` names. Segment 1 is `1000`, segment 2 is `1008`, and the usual step is 8, with gaps.	`TBdecompile/maps/rtm-ne-to-ghidra.md`

7.1 SoftICE

The kit is `SI405w9x.exe` plus ICEDUMP 6.021. The loader that worked was `w9x/4.05.334/icedump.exe`. It loaded on the Windows 98 VM after both `kernel32.dll` and `user32.dll` exports were listed in `winice.dat`.

Ordinal breakpoints are stable across boots. Selectors are not. `TB80BAS.Ordinal_8` is `CDBFILEOPEN` on every boot. `36F7:03DE` is `FUN_1160_03de` only until the next reboot.

`/DUMP` writes the SoftICE console, not the address given, unless the dump experts are set. Several early `page.bin` and `rtm.bin` files are command logs or the DOS device chain (`CON`, `AUX`, `PRN`). A dump is real when it contains book strings at the file offsets already known, as `page.bin01511` did (`ASYM_TpID`, `Page#`, matching `9e954d01.tbk` at `0xB8B51`).

Do not mix these commands into a WDW session.

7.2 WDW

On the Watcom debugger:

- Create the breakpoint first, then set `resume on`.
- Qualify with `@`, not `!`.
- Log with `log .`
- There is no trace that prints every call. The three logs in `tbwp0026` are named-export hits: 11 in W16, 53 in UTL, 173 in BAS.

Those three logs are described in chapter 4. They do not contain the reader. The next useful break is after `CDBFILEOPEN`, on the lesson path, not another pass over the same exports.

GPF notes from the SoftICE sessions still apply in spirit: do not step into an unloaded segment, and do not single-step a far call into `KERNEL` unless that segment is present. The CMP expand host died in `TB80W16` at `001c:00000015` (`9A 9C 06 57 2A`, segment not present). That fault is parked. It is not the book-open path.

7.3 Unicorn

`unfold.py` enters 16-bit protected mode, calls `BootApp`, and dumps each segment `LoadAppSeg` materializes. Pass 01 filled 3 of 139 RTM segments. Pass 02 filled 138, after the call stub was flushed from the Unicorn cache so every call did not replay segment 2. Passes 03 through 07 bind imports, including 461 ToolBook ordinals to real DLL segments. 339 imports stay

Windows stubs. The host still dies at `FFFF:0158` and `0080:0098` before a DOS open of the book.

Thought (WV-008). A real Win16 return is what would get that host further. The three hosts (the RTM experiment, `newl`, `neww`) are not one program and do not share a Windows table (WV-035).

The path that does open a page is the pipeline host, which parses the file instead of running RTM. On `9e954d01.tbk` that open is at `0x9EC`, 129 pages. `render.html` is written only when the debug-render flag is set (WV-034).

7.4 Password cipher

`TBCBT` segment 7 (Ghidra `Code7`, base `1030`) holds the four cipher exports. A SoftICE module list that showed segments `01`, `0E`, `02`, `0B`, `08`, `0D`, `04`, `03` had not loaded segment 7. The dialog is an ordinary Windows dialog. The cipher is not in the dialog procedure.

After a failed password, the typed string and the constants `0xA01` and `0x300` remain in memory. Searching for those constants finds the call site even while segment 7 is unloaded. The three addition schemes, the with-password file delta, and the fact that the stand-in key does not decrypt it are chapter 16. The NIDA `.sbk` passwords are not recorded here. They are not required to read the decompile.

7.5 Ghidra

Two NE loader bugs were fixed for this work: the entry-table segment byte was treated as signed, and `ne_cbenttab` was ignored. `TB80BAS` is the image with entry segment bytes at or above 128. The unpacker does not rewrite those bytes. The Ghidra patch does. An honest BAS image loads in the patched Ghidra 12.1.4 build.

Force-disassemble scripts recover functions the loader never reached. One walks undefined `C8` (`ENTER`) bytes. The next accepts a real `RET` / `RETF` before treating the following byte as code, and it only creates functions at entry-like symbols. Run `C8` first, then the prologue pass. They do not name anything, and they do not touch code that is already defined.

`SENDMESSAGE` (321 hits in one RTM listing) and `LSTRLEN` (282 hits) have mangled arguments. That is a decompiler problem, not a missing function. It wants a fresh decompile, not another reading of the same HTML.

8. Naming

Claims

Id	Grade	Statement	Source
N-001	Known	Eight modules were counted for pattern coverage: 10,991 unique functions, 1,918 nameable by pattern, 17.5 percent.	`toolbook_naming_progress.md`, 2026-10-11
N-002	Known	Confirmed renames, after the later digs, are 112: BAS 20, then BMP, RTM, CMP, UTL, W16, LNL, MM, RCR, FLT, NET, JPG, and VBX.	`naming_summary.md` through 101, then JPG +1 and VBX +6
N-003	Known	A pattern name is not a confirmed name. The nameable lists are candidates. The rename scripts are the ones that were applied.	`TBdecompile/rename/`
N-004	Known	Coverage by module, pattern only: RTM 32.7 percent, UTL 41.1, BMP 26, BAS 17.4, W16 17.3, LNL 12.2, CMP 5.6.	progress table

8.1 What the numbers mean

“Nameable” means the decompiler text matches a pattern: a GDI call, `GLOBALALLOC`, `SENDMESSAGE`, a string move. “Confirmed” means a person or a script renamed that one function because the body matches the name. The gap between 1,918 and 112 is mostly unreviewed candidates, not unknown programs.

The 30 percent target in the October 11 note was a counting goal, not a research goal. Naming another thousand `StrCpy` wrappers will not find the record reader.

8.2 Confirmed painters and neighbors

These are the names later chapters use. Addresses are Ghidra `segment:offset` on the image those scripts were run against. Selectors move. Offsets inside the NE segment do not, once the segment is identified.

Name	Address	Module
`PaintPolygon`	`1408:0376`	BAS
`PaintPolyline`	`13e8:00E6`	BAS
`PaintRectangle`	`1020:03E4`	BAS
`PaintArc`	`10A0:0558`	BAS
`PaintEllipse`	`1230:2F1C`	BAS
`PaintRoundRect`	`1230:2278`	BAS
`ScaleTwipsToScreen`	`1008:01DD`	BAS
`SetProperty`	`12D0:03EC`	BAS
`SetPointArray`	`1408:00D8`	BAS
`SetupPaintRect`	`12D0:086C`	BAS
`SetScriptProperty`	`1008:03F6`	BAS
`Lz77Expand`	`1000:0031`	BMP
`CreateDibBitmap`, `BitBltBmp`, `SelectPaletteBmp`	BMP rename script	BMP
`LoadCompilerResource`, `ExpandTextCore`	CMP rename script	CMP
`PaintPageRect`	`1398:05C6`	RTM

Ellipse, round-rect, and a second arc painter were added in the same BAS pass. They are real GDI wrappers. They are not evidence that every shape in a lesson is stored as that GDI call’s arguments. Chapter 5 is the rule for storage.

8.3 Scripts

Java and Python rename scripts live in `TBdecompile/rename/`. Copies of the nameable TSV files live under `TBWorldview/Modules/tb80*_knownnames/`. The scripts rename only when the old `FUN_` name is still present. They do not invent names for functions that do not match the body.

Ghidra `CodeN` is not the NE segment number. Use `TBdecompile/maps/` before typing a segment into a debugger.

8.4 What is still thin

`TB80RCR` exports are only partly named. The `TB80UTL` classification strings are known. They are not yet tied to the kind bytes inside `14 00 00` chunks. CMP is 5.6 percent pattern-nameable because most of it is the compiler, not a wrapper around a Windows API. A name on a CMP function is useful when it is `COMPILEREXPANDTEXT`. A name on a CMP function that only moves a token is not.

9. The modern stack

Claims

Id	Grade	Statement	Source
M-001	Known	The clean-room reader is not the engine. It does not score quizzes, place every bitmap, or run the NE images.	WV-005
M-002	Known	Foolbook opens `.tbk`, `.sbk`, `.fbk`, and `.fsk`. Shapes come from kind `0400` polygons and from bounds-only rectangles.	Foolbook `engine/shapes.py`
M-003	Known	Foolcourse is the process that serves a book. Foolbook is a local window. If Foolbook is started by a course, it pulls the book and skips the splash.	release notes, 2026-10-13
M-004	Known	The dependency order is Foolversity, then Foolcourse, then Foolbook, then `openscript_analysis`. Foollege is the layer under Foolversity that holds courses. It is not a running server in the current release.	hierarchy note
M-005	Known	TBpipeline 1.0.0r4 parses `9e954d01.tbk` (129 pages) without running RTM. A known bug is an 18-versus-23 render count.	pipeline release note
M-006	Thought	A reader that prefers explicit points, then TBX Vertices, then bounds, will look like the lesson often enough to study it. It will not be a substitute for the Win98 runtime until the reader in chapter 4 has a name and the picture id is known.	project philosophy

9.1 Foolbook

The engine parses the 8.1 container. It does not load `TB80BAS`. Polygons use the kind `0400` layout (count at `+0x1A`, pairs at `+0x22`). Rectangles use the bounds-only tail (`01 00 00 04`). On Logic Families that is on the order of 165 shapes from the early polygon-and-rect pass, and 961 rectangles from the later marker scan. Both counts can be true. They are different patterns.

System books: core `.fsk` first, user `.fsk` after, as mods. `.sbk` is accepted. It is not required for the v1 player.

The window is native. A direct start opens `using_foolbook.fbk` and a splash that asks whether to resume the last page. A start from Foolcourse, or a start on a named file, skips the splash and uses the last page the course recorded. The splash text is a joke (`2+2=5` / “For greater values of 2.”). It is not part of the book format.

OpenScript in this player is the subset in chapter 6. ActiveX, Python, and C++ as replacement engines are a stated direction. They are not implemented.

9.2 Foolcourse and Foolversity

Foolcourse is the only server in the current release. Foolbook does not listen. The earlier idea that Foolbook itself served on port 1337 was withdrawn: a local viewer should not be the thing that stays up after the tab closes.

Foolversity writes students, schools, and curricula. Foolcourse reads them. The hierarchy, as stated, is: a Foollege sits under Foolversity and has Foolcourses; a Foolcourse has Foolbooks; a Foolbook can run OpenScript. Foollege is not the current server. A later Foolcollege was discussed as the place that would hold several courses for several students. That program does not exist yet.

Install is a virtual environment and `pip install` of the Foolversity git URL. A system Python on Debian will refuse the install. That is PEP 668, not a missing package. If a dependency has not changed, a reinstall should not download it again unless a fresh-start flag is set. That rule was written after several reinstalls pulled `openscript_analysis` for no content change.

9.3 TBpipeline and the unpacker

TBpipeline’s useful stage is the file parser: load order recorded from a SoftICE log, then a TBK open, then a page walk, then HTML. The Unicorn RTM stage still stops before the book is opened, so a render from this package is a parse, not a playback. Curves, including an attempt at the P420 flame, are emitted as separate files. They are not placed on the page (WV-004’s falsifier still stands for placement).

`toolbookunpacker` produces an honest NE image. It clears the Optloader self-load flag. It does not patch the entry table for Ghidra. That patch lives beside the unpacker as an optional post-step. Only BAS, on the current set, has entry segment bytes at or above 128.

9.4 What “getting NIDA working” means

Three different finishes are easy to mix up.

Finish	Done when
A page can be read	Field text, jumps, and the decoded shapes for that page match the Win98 screen closely enough to study the lesson.
A lesson can be walked	Next, Back, menus, and quiz flow follow the handlers, including the jumps into `nida81.tbk`.
A lesson can be run	The system books load, `initializebase` / `initializerecs` / `initializetest` do what the DLL exports imply, pictures land on their widgets, and the unnamed reader is no longer unnamed because the clean-room code does the same thing for the same bytes.

The first is partly done for Logic Families. The second is partly done for the menu and the quiz ids that have been transcribed. The third is not done. Authorship and a modern `TB80CMP` are side work. They help the next person. They do not open P1040.

10. Open, retired, and superseded

10.1 High

These block a faithful page.

Gap	Grade	Why it is still open	Next step that would close it
Runtime record reader	Thought that it is in BAS after `CDBFILEOPEN`, before first paint. No name.	Every static candidate in chapter 4 was something else. The WDW export traces do not contain it.	Break after `CDBFILEOPEN` on a lesson open. Dump the buffer that is about to become an object. Do not break ordinal 51.
Picture id	Known that BMPs are in the file. Not known which property binds one to a widget.	67 markers paired with 72 TBX bitmaps, and the widget still has no name for the bitmap.	One live `TB80BMP` ordinal 3 stop, with the object pointer, compared to the page blob.
Property expansion	Known that `FUN_12c8_0ef8` copies four words from a structure that is not the disk record.	Offsets `+0x3E` and friends do not match a `14 00 00` header.	Only useful on the author path. For runtime, fold this into the reader break. Do not keep it as a separate `CDBSENDCREATE` hunt.
Tbload vtable	Thought. `FUN_00406a70` calls vtable+4 after the marker search.	The 32-bit search is found. The callee is not.	One break at `00406aa1` with the book path in a register. Low value until the reader is the goal again.
Polygon type table	Thought. Built at runtime.	No static pointer to `PaintPolygon`.	Break `PaintPolygon`, walk one frame up, dump the table.

10.2 Medium

Gap	Grade	Note
P420 stroke breaks	Thought	Three 96-point groups in TBX. One tangled dump from `+0xA13`. Look for length prefixes between runs before inventing a new curve format.
Arc on disk	Known live, unknown on disk	Type `0x11`, four pairs, fill at `+0x18`. Angles not located. No `GDI!PIE`.
Stroke color and pen width	Open	Fill is known. Segment `1018` creates pens and does not say who frees them.
Palette on an 8 bpp BMP	Known that segment `1150` builds a palette	Not known which palette a given carved BMP must use.
`DRAWTEXT` beyond three offsets	Open	`param_5+0x4A`, `+0x4C`, `+0x50`.
Per-pin index `+0x52`	Known the field exists	Not applied. Shared point bytes stack.
Background, parent, z-order	Known the TBX id is not a word in the page	P1170 bracket is parent-relative. The parent link is not decoded.
`initializebase` and `toolbook.lst` order	Thought	Names are in the lesson. The call log is not.
OpenScript keyword lock	Thought	Instructor 8 compile, then expand. Not run.
`SENDMESSAGE` / `LSTRLEN` arguments	Known mangled	Re-decompile. Not a new algorithm.

10.3 Retired

Idea	Grade	Why it stays closed
On-disk `14 00 00 01` becomes `14 00 00 11` in memory	Not true	Both values exist on disk. A later live object used `14 00 00 02`. WV-010.
`FUN_1028_2680` is the reader	Not true	Filter registrar.
`FUN_1008_2d31` is the reader	Not true	Math.
RTM contains the reader	Not true	Paint handoff only.
`CDBCHUNK*` is the reader	Not true	Text separators.
`GDI!PIE` draws pies	Not true	No import.
The reader must contain the literal bytes `14 00 00`	Thought, not a requirement	Length-prefixed walks would not. Do not close a function for lacking the literal, and do not keep scanning for the literal as the only method.
HTML listing absence	Not evidence	`??` runs and forced-disassembly passes exist.

10.4 Superseded notes

Read these for the function bodies. Do not read them for the runtime path.

Note	What it got wrong
`tbworkingpapers/tbwp0002/bas-create-path.md`	Put `CDBSENDCREATE` on book open.
`tbworkingpapers/tbwp0008/cdbsendcreate-sites.md`	Same chain, more callers. The callers are author-side.
`TBdecompile/20261015/00-how-toolbook-operates.md` §2	Same chain, written as “the create path that has been identified.”
`TBdecompile/20261015/01-theories-of-operation.md` §1	Still calls the reader “the caller of `CDBSENDCREATE`.”
`TBdecompile/20261014/02-still-open.md`	Same wording. The rest of that file’s open list is still the open list.
`TBWorldview/ToolBook_Architecture.md` book-open flowchart	Draws `CDBSENDCREATE` once per widget.

The correction, in one paragraph: Instructor calls `CDBSENDCREATE` when it creates or rewrites an object. A lesson open does not. The function that turns a `14 00 00` chunk into a live object during that open has not been named. It does not call ordinal 51. `FUN_12c8_0ef8` is on the author side of the same line.

10.5 Side quests that are real and not the reader

Kept because they will matter after the reader has a name.

- Object pool: `FUN_1478_014a` allocates near objects. `FUN_1478_00a0` is the pool.
- RTM dialog chrome: `GETDLGITEM` id `0x65` (55 hits), `GETWINDOWLONG` index 10 (196 hits). `FUN_1068_0098` is `GLOBALALLOC` with an out-of-memory retry.
- Author mode exports on BAS, and `GETAUTHORLOCALIZEDDLL` on RTM. The author binaries themselves are not in the archive.
- Password cipher in TBCBT segment 7. Constants `0xA01` and `0x300` after a failed password.
- CMP expand, parked at the W16 segment-not-present fault. Not on the NIDA open path.

10.6 Order of work

1. One lesson open under WDW. Break `TB80BAS@CDBFILEOPEN`, resume, then break the next BAS export that runs before the first paint. Log it. 2. When an object exists, dump `+0x0F` and the page offset it came from. That pair names the reader, whatever the export happens to be called. 3. Only then place bitmaps, split the P420 strokes, and apply `+0x52`. 4. Do not schedule another full-module naming pass, another Unicorn boot, or another break on ordinal 51 in service of step 1.

10.7 The loose sheets are not a second record

Chapter 18 names every markdown sheet under `tbworkingpapers/` and the section of this directory that replaced it. A sheet can still describe a real function and still be wrong about when that function runs. Leaving the sheet on disk keeps the trail. It does not keep the old claim alive. The printable form of the claim is the PDF next to these chapters.

11. Keepers

Addresses in this chapter are from the 2026-10-15 Windows 98 boots, DGROUP `347F`. Match a later boot on the `enter` size and the next two or three instructions, not on the selector. Decompile line numbers are from `TB80BAS.DLL.c` and do not move when the selector does.

`CDBSENDCREATE` is not in this list. It is author-only. See chapter 4.

11.1 File open and close

Function	Live (one boot)	Enter	Decompile line	Role
`CDBFILEOPEN`	`4C7F:017A`	`000A`	27592	Book-open export. Ordinal 8.
`CDBFILECLOSE`	`4C7F:0948`	`0002`	27965	Close. Same caller cluster as open.
`CDBFILEPATH`	`4C7F:09E0`	`0002`	28002	Path query.
`CDBFILESTATUS`	`3ACF:0000`	`0004`	30210	Status.
`CDBQUERYMULTISAVE`	`4C7F:0B90`	`0018`	28108	Multi-save query.
`CDBMULTISAVE`	`4C7F:0E24`	`000C`	28272	Multi-save.
`OBJFILEOPEN`	`3147:177C`	`025A`	76162	Large frame. Buffered read. Downstream of the reader.
`OBJFILECLOSE`	`3147:10EA`	`0002`	75887	Object-file close.
`OBJGETFILE`	`3147:0094`			Object file access. Seen on the page path.
`OBJGETHDRLONG`	`3147:0242`			Header long. A 4-byte read here was an internal pointer, not a page header.

`CDBFILEOPEN` callers seen: `322F:3252` (jump), `3D6F:0416` / `0F79` / `0E8E`, and the `3247` spine. `OBJFILEOPEN` callers include `3D6F`, `3247`, and `OBJGETFILE`.

11.2 Object lifetime

Function	Live	Enter	Decompile line
`OBJFREE`	`3147:0CCC`		75619
`OBJFREENEAR`	`3147:0D70`		75662
`OBJUNLOCK`		`004A`	call site, not a definition
`OBJREALLOCNEAR`	`3147:08D6`	`0018`	
`CDBCLEARBOOKREFS`	`4C7F:10A6`	`00B8`	28451
`CDBINIT`		`00C8`	30304

`HEAP*` / `TBHEAP*` hand out the `0x76`-byte object. `TBHEAPCLEARCRITSECT` was the last stop before a window appeared on one boot. `TBHEAPALLOC` and `TBHEAPFREE` are call sites throughout the decompile. There is no standalone `TBHEAPCLEARCRITSECT` definition in that listing.

11.3 Page, instance, and the value bus

These are closer to the unnamed reader than ordinal 51. On the 2026-10-15 captures, `CDBSENDCREATE` was listed in the Functions window and did not appear in the Calls stacks. `CDBCREATEPAGEREF` and `CDBSETVALUEEX` did.

Function	Live	Enter	Decompile line	Note
`CDBCLASSTABLEGETBEHAVIOR`	`44F7:121A`	`0008`	125154	Splash gate. Unbreaking it let the splash run.
`CDBCREATEINSTANCE`	`480F:0000`	`0002`	24974	Instance constructor.
`CDBCREATEPAGEREF`	`483F:03B2`		25913	Page reference. Next break.
`CDBVIEWNONE`	`483F:02F8`	`0002`	25877	View state.
`CDBVIEWPAGE`			25726	Ordinal 15.
`CDBVIEWBKGND`			25825	Ordinal 18.
`CDBSETVALUEEX`	`3397:147C`			Extended property set. Next break.

Function	Live	Enter	Decompile line	Note
CDBGETVALUE	3397:0D54		38475	Property bus.
CDBGETVALUEEX			38488	Extended get.
CDBPARENTVALUE	3397:07B2			Parent value.
CDBDEREFVALUE	3397:0400			Deref.
CDBLOCKVALUE	3397:02CA	0002	37826	Shifts 0B[bp]. Not a record parser.
CDBCLIENTSTATUS	480F:04C6			Client status.
CDBREGISTERINSTWINDOW	480F:0522			Instance window.
CDBDISPLAYSTATE	4877:0000			Display state.
CDBSYNCSCREENS	4877:0276			End of the author create path. Seeing it does not mean ordinal 51 fired.
CDBSENDSELECT	44CF:009E			Select. Not create.

Value constructors in segment 4267 fire often and are not the stream walker:

Function	Live	Enter
CDBVALUENEWBOOKREF	4267:0000	0004
CDBVALUEBOOKFROMSTRING	4267:03F4	0018
CDBVALUELAYERFROMSTRBOOK	4267:05F8	004C
CDBVALUERESOURCEFROMSTRING	4267:08B2	004A
CDBVALUEGRAPHICFROMSTRING	4267:0AA0	0074
VALUENEWBOOK	4267:0D90	0006
VALUENEWRESOURCE	4267:0EFC	000E
VALUENEWGRAPHIC	4267:1494	0008
CDBVALUENEWGRAPHICREF	4267:02D4	
CDBVALUENEWLAYERREF	4267:00C2	

VALUENEWLONG builds a long. It is not a record parser.

11.4 Chunk walkers

These walk characters: comma, CR, LF, tab, space, quote. A decompile that shows ISDBCSLEADBYTE and a comma test is this family.

Function	Live	Enter	Decompile line	Callers
CDBCHUNKEND	0D37:0000	0004	138456	3187:0210, self from FIND
CDBCHUNKFIND	0D37:0146	0008	138575	4857:0C68, 3187:0210, 322F:0577
CDBCHUNKLAST	0D37:0334	0010	138676	4857:090B, 3187:0210

11.5 Paint and resource binding

Function	Live	Decompile line	Note
CDBPAINTCLIENT	389F:0506	13056	Ordinal 26. RTM WM_PAINT ends here.
IMBITBLT		13277	BitBlt path.
IMDRAWLEVELS		13353	Draw levels. IMDRAW as a bare name is not a definition.
IMBLACKOUTBMP		13995	
IMINVERTBMP		14021	
IMSHOWBMP		14047	
RESMGREXTRACT		21513	Resource manager.
RESMGREXTRACTGDI	4D27:1730	21523	Extract for GDI.
GETDIBCOLORCOUNT	35D7:053A		Enter 0006.
GETDIBSIZE	35D7:05DA	not in the BAS or BMP listing	Enter 0008.

Function	Live	Decompile line	Note
HUGEGETDATA	33BF:0D6C		Enter 0018. Large fetch.
COLORHLSTORGB	44B7:0B84	155991	Menu color. Enter 0034. Not the page painter.

Painter offsets that do not move inside the segment are in chapter 8. PaintPolygon is BAS segment 0x82 offset 0x376.

11.6 Incidental on open

Seen, and not the reader: CDBREGISTEROLE2WINDOW (3767:0000), CDBREGISTERVBXFORM (4437:0000), VBXNEWVIEW (4437:0216), VALUEARRAYINFO (4357:02F2), VALUEARRAYSET (4357:0932), VALUEARRAYGET (4357:05D8), HEAPVALUETOPOINTER (450F:0F28), HEAPVALUETOFRAME (450F:0F4E), HEAPENTERCITSECT (450F:0DCE), HEAPLEAVECRITSECT (450F:0DF6).

?SENDPROC@TBXBOBJ@@ was seen at 29AF:0000 (cmp ax, 3014). The decompile does not contain SENDPROC@TBXBOBJ as that name. Related symbols are _SETPROC_TBXBOBJ at line 2175, _SENDPROC_TBXLAYER, _SENDPROC_TBXBASE, and _SENDPROC_TBXUSER.

12. The book-open spine

12.1 The frames that repeat

On the 2026-10-15 book-open and page walk, these frames were in nearly every WDW Calls window. Selectors are that boot only.

Selector:offset	Note
`3497:03EF`	Early frame.
`3497:4A6E`	The planned step point.
`3197:0379`	
`3167:0CB1`	Near call.
`3167:14BC`	Far call.
`3247:02BF`	Near.
`3247:05B7`	Near.
`3247:0B7A`	Far.

Everything else in those captures hangs off this spine. The full selector-to-decompile table for that boot is `maps/wdw-to-ghidra.md` (485 lines). It is a lookup, not a finding. Do not treat a selector from that file as valid on the next boot.

12.2 What the three export traces contained

WDW has no full call trace. `break resume on` plus `log` logs named exports and continues. `resume on` is a property of a breakpoint that already exists. Module qualification is `TB80BAS@OBJFILEOPEN`. SoftICE `!` does not work in WDW.

Trace	Shots	What fired	What did not
W16	11	`GETPLATFORMREF`, `IS32BITENV`, `GETRUNTIMELOCALIZEDDLL`, `ACXINITIALIZETOOLBOOKSUPPORT`, `ACXOLEINITIALIZE`, `ASYM_STRINGFROMCLSID`, `SETTHUNKINGLAYERSERRORFUNC`, `LOADLIBRARYEX32W`, `GETPROCADDRESS32W`	Book open. Record walk.
UTL	53	`GETCHARINFO`, `GETSTRMGRSTRING`, task collections, `PLUGINSVCINIT`, `LOADRESSTR`, `CXTHELPINIT`, `GETTBKINIPROFILEINT` / `STRING`, `INITMETRICS`, `SETCONVERSIONDC`, `CXTTOPXT`, `GHEAPCREATE`, `GHEAPALLOC`, `SYMLOOKUPNUMBER`	Record walk. The type-name table lives here and was not the thing being stepped.
BAS	173	`VALUETOSTRING`, `VALUENEWSTACK`, the `VALUENEW*` family, `OBJFILEOPEN` (`enter 025A`), `STACKGETSIZE`, `STACKPUSH`, `CDBCHUNKLAST`, `CDBCHUNKFIND`, `CDBCHUNKEND`, `RESMGR*`	A compare of a far pointer to `0x14`. Ordinal 51.

`OBJFILEOPEN` in that BAS trace is the file-open path. The buffered reader is `FUN_1160_03de`, reached through `FUN_1258_00a8`. When the buffer satisfies the request, the raw DOS path at offset `04F3` is skipped. Live 4-byte reads returned an internal pointer (`FC B5 BD B0` in one dump), not a page header and not `03 4A 42 4F`.

12.3 Where to break next

1. `TB80BAS@CDBFILEOPEN` on a lesson open. Resume. 2. Then `CDBCREATEPAGEREF` or `CDBSETVALUEEX` (export names, not the selectors above). 3. When an object exists, dump `+0x0F` and the page offset it came from. That pair names the reader. 4. Do not break ordinal 51. Do not schedule another pass over the same W16 and UTL exports in order to find the reader.

The reader is still unnamed. The working model is in chapter 4: a length-prefixed walk in BAS after `CDBFILEOPEN` and before the first paint, with no literal `14 00 00` compare required.

12.4 Debugger facts that keep wasting a session

SoftICE field work closed 11 October 2026 (`SI405w9x` plus ICEDUMP 6.021). Ordinal breakpoints survive a reboot. Selectors do not. `TB80BAS.Ordinal_8` is `CDBFILEOPEN` on every boot.

`/DUMP` writes the SoftICE console unless the dump experts are set. Several early `page.bin` and `rtm.bin` files are command logs or the DOS device chain (`CON`, `AUX`, `PRN`). A dump is real when it contains book strings at offsets already known, as `page.bin01511` did (`ASYM_TpID`, `Page#`, matching `9e954d01.tbk` at `0xB8B51`).

Symbol maps are `NE-segment:offset` names (`TB80BAS.map`, 3799 entries; `TB80RTM.map`, 2669). SoftICE and WDW do not load a `.map` as a module.

The CMP expand host died in `TB80W16` at `001c:00000015` (`9A 9C 06 57 2A`, segment not present). That fault is parked. It is not the book-open path.

The Unicorn host has not opened a book. Passes 01–07 fill segments and bind 461 ToolBook ordinals, then die at `FFFF:0158` and `0080:0098`. Pass 06 exits after `TB80.INI`. The three hosts (the RTM experiment, `newl`, `neww`) are not one program.

13. Specimen pages

Every number in this chapter is from `9e954d01.tbk` (Logic Families) or from its ToolBook 11.5 Save To XML, unless a path says otherwise. Screen pixels are stored twips / 15 at 96 dpi.

The binary has 129 pages, each a record starting `80 65 00 05 00 02 02`. The export has 131 `Page.xml` files. P440 and P2920 have no `80 65` record. They are an export addition. All 129 binary names match a TBX page.

TBX folder names (`Page880`) are export ids, not the runtime page name (`P1040`). IdNumber 880 is page `P1040`.

13.1 P1040 — two DIP packages, bounds only

File offset `0x19C03A` (1687098). Page blob 7764 bytes. Background id 2. Next goes to `P1050`. Back goes to `P1001`.

The 8.1 file does not store the 11-point pin tips that the XML writes. Each pin is four twip words, a zero dword, fill `0xC0C0C0`, then marker `01 00 00 04`. The tip is expanded at paint, and the export writes that expansion as Vertices. The book has 961 copies of the `01 00 00 04` marker. P1040 uses 28 of them for pins and 2 for bodies.

7400, right package

Body at blob `0x09F4`: bounds `4875, 3555, 7830, 4605`, screen about `325,237–522,307`, fill `64,64,64` (`0x404040`). Label field text `7400`.

Bottom pins, top `4605`, bottom `4860`:

Blob offset	Bounds (twips)	Screen about
`0x03E4`	`5055–5205`	`337,307–347,324`
`0x044C`	`5460–5610`	`364,307–374,324`
`0x04B4`	`5880–6030`	`392,307–402,324`
`0x051C`	`6315–6465`	`421,307–431,324`
`0x0584`	`6705–6855`	`447,307–457,324`
`0x05EC`	`7125–7275`	`475,307–485,324`
`0x0654`	`7560–7710`	`504,307–514,324`

Top pins, top `3300`, bottom `3555`:

Blob offset	Bounds (twips)	Screen about
`0x071C`	`5055–5205`	`337,220–347,237`
`0x0784`	`5460–5610`	`364,220–374,237`
`0x07EC`	`5880–6030`	`392,220–402,237`
`0x0854`	`6315–6465`	`421,220–431,237`
`0x08BC`	`6705–6855`	`447,220–457,237`
`0x0924`	`7110–7260`	`474,220–484,237`
`0x098C`	`7530–7680`	`502,220–512,237`

The first pin's bounds `5055, 4605, 5205, 4860` are the start of the TBX Vertices list. That is the expansion direction: bounds in the 8.1 file, point list in the XML.

5400, left package

Body at blob `0x117C`: bounds `1500, 3555, 4455, 4605`, screen about `100,237–297,307`, same fill `64,64,64`. Label `5400`.

Bottom pins, y `4605–4860`: `0x0B6C 1680–1830`, `0x0BD4 2085–2235`, `0x0C3C 2505–2655`, `0x0CA4 2940–3090`, `0x0D0C 3330–3480`, `0x0D74 3750–3900`, `0x0DDC 4185–4335`.

Top pins, y `3300–3555`: `0x0EA4 1680–1830`, `0x0F0C 2085–2235`, `0x0F74 2505–2655`, `0x0FDC 2940–3090`, `0x1044 3330–3480`, `0x10AC 3735–3885`, `0x1114 4155–4305`.

TBX group ids: 53 is the 7400 package, 52 is the 5400 package, 58 is the discussion title (gold plate button, graphic "bitmap id 149", plus a white "DISCUSSION" field and a blue drop shadow). The widget still does not name that bitmap inside the 8.1 object. See chapter 5.

Intro field bounds `600, 600, 8805, 2490`. Next button `8907, 6210, 9342, 6630`. Back button `-3, 6210, 432, 6630`.

13.2 P430 — kind `04 00` polygons, including a live pin tip

`9e954d01.tbk` has 96 kind-`0400` records on 13 pages. P430 has 22. A search that ignores the kind word produces false positives.

Record layout: fill dword at `+4`, kind `04 00` at `+0x0C`, count word at `+0x1A`, twip pairs at `+0x22`. The painter is `FUN_1408_0376`. It copies object `+0x30` / `+0x32` to `DS:30AC` and calls `GDI!POLYGON`.

P430 draws as a chip: silver pins (`0xC0C0C0`) and a light-gray body (`0xE0E0E0`). One live 11-point polygon is a pin tip, not the body. Captured screen points ran `(218,335)` through `(218,341)`, a 10 by 17 notch. The same points are in the file as twips near page offset `0x9CB`: `(3147,3862)` through `(3147,3973)`.

Pins that share identical point bytes stack until the index at object `+0x52` is applied. The clean-room draw does not apply it.

13.3 P420 — the curve that is not yet a flame

Points sit at page `+0xA13`. Curve's type id in `nc81base.sbk` is `0x05C7` (file offset `0x1810`). That id is not in the P420 page records. A naive walk draws a tangle in screen space around 400–500. The TBX export is three 96-point red IrregularPolygons. There is no `Curve` name on the page.

Thought. The tangle is strokes concatenated that should be drawn separately. A length prefix or a zero terminator between groups has not been proven. Do not treat a raw dump of `+0xA13` as the flame.

13.4 P1170 — the bracket is computed, and the word table is an index

Live polyline: `(354,201) (321,201) (321,238) (358,238)`. Object bounds in twips: left `4801`, top `3015`, right `5370`, bottom `3570`. Divided by 15 that is the same rectangle. BAS segment `0x5B` computes the four points. They are not stored.

The 27 ascending words at page `+0x8C3` are an offset directory into a `14 00 00 83` blob, not a Group. On this page the blob is file `0x1AB651`, 1844 bytes, and the word table is file `0x1ABE97`. The words begin `0x14, 0x16, 0x40, 0x52, 0x66, 0x7A, 0xC2` and are relative to the blob. Script-like markers inside the blob (`buttonClick`, `"P1180"`, `"P1140"`) sit at blob `+0xA9`, `+0xB9`, `+0x17B`, `+0x18B`. After those regions are set aside, the table still points at short structured runs (type bytes, small integers, `0xFFFFFF`). A Group is its own object and has its own member list.

13.5 P400 record sizes

P400 is 7046 bytes and 12 records. The early walk, kept because the sizes show what a "page of widgets" actually is:

#	Page offset	Length	Kind	Note
0	`+0x2B`	33	`0001`	Small.
1	`+0x4C`	31	`0000`	Small.
2	`+0x6B`	18	`0D00`	Small.
3	`+0x7D`	1458	`D900`	Content blob: scripts and colors.
4	`+0x62F`	2828	`0021`	Large. Not a polygon.
5	`+0x113B`	894	`0000`	Count word at `+0x1A` is 21, not a kind-`0400` polygon.
6	`+0x14B9`	156	`1600`	Small.

13.6 Lesson jumps that have been read

Page	Handler
P900	TTL → P1001. Other → P2001. Final quiz → P10001. Exit → `nida81` P20001. No Next.
P1001, P2001	Back returns to P900, not to `nida81` P20001.
P300	`selection1` sends `lastbook nida81.tbk` and `nextpage` P310.
P10001	Directions → `nida81` P10020. Exit → `nida81` P20001. Begin → P10210.
P10410	Menu returns to P900.
P1290	Correct choice 5408. Again → P1292. Next → P1300.
P20000 of `nida81.tbk`	`buttonClick` goes to P20001 because the handler says so.

13.7 TBX draw inventory

Tag counts across the `9e954d01` export. These are XML tags, not kind-`0400` records. A page can have hundreds of IrregularPolygons that the 8.1 file stored as bounds.

Kind	Count
IrregularPolygon	654
Line	528
Field	497
Button	382
Group	371
Rectangle	243
AngledLine	66
Arc	62
Ellipse	45
Hotword	5

Pages whose export folder contains at least one IrregularPolygon. Columns are IrregularPolygon, Rectangle, Line, AngledLine, Arc, Ellipse, Group, Field.

TBX folder	Poly	Rect	Line	Angled	Arc	Ell	Group	Field
Page872	14	2	2	0	1	1	4	7
Page873	44	5	0	0	2	2	2	4
Page874	8	2	2	18	0	0	2	2
Page875	96	8	2	1	8	0	8	1
Page876	12	32	108	0	0	0	48	5
Page880	28	2	0	0	2	0	5	5
Page885	28	4	0	0	2	0	4	5
Page887	14	1	0	0	1	0	2	8
Page891	14	1	0	0	1	0	2	8
Page897	14	1	0	0	1	0	2	8
Page900	14	1	0	0	1	0	2	8
Page901	42	3	0	0	3	0	3	4
Page909	14	1	0	0	1	0	2	8
Page912	42	3	0	0	3	0	5	7
Page914	28	4	0	0	2	0	5	6
Page915	42	3	0	0	3	0	5	7
Page917	42	3	0	0	3	0	4	8
Page918	42	3	0	0	3	0	4	8
Page919	14	1	0	0	1	0	1	4
Page923	28	2	0	0	2	0	5	5
Page931	3	8	27	0	0	0	13	4

TBX folder	Poly	Rect	Line	Angled	Arc	Ell	Group	Field
Page932	9	28	81	0	0	1	37	5
Page934	14	1	0	0	1	0	2	2
Page940	6	22	54	0	0	2	24	5
Page941	14	1	2	0	1	0	2	4
Page943	14	1	2	0	1	0	2	4
Page949	2	0	7	0	0	0	0	2
Page961	12	32	108	0	0	0	49	7

Page880 is the P1040 export: 28 polygons, 2 rectangles, 2 arcs. The 28 polygons are the paint-time expansion of the 28 bounds records in section 13.1. Page875 (96 polygons, 8 arcs) is the dense drawing page in the export. It is not, by itself, proof of 96 kind-`0400` records.

13.8 Pictures, stated with the two offsets kept apart

Windows BMPs can be carved. One match against the export paired 67 markers with 72 TBX bitmaps. The widget does not name the bitmap. Compressed pictures go through `TB80BMP` ordinal 3 and have been followed as far as a far call at `0x1FCD`. `Lz77Expand` is BMP `FUN_1000_0031`.

The shared near-start graphic is a 581 by 157, 8 bpp gold plate, not a text banner.

Two notes locate the `nida81` splash and they are not the same byte: file `0x413FAB` (323 by 344, 8 bpp, off-bits 310), and a carved copy taken from file `0x487BE` (27874 bytes). The live `BITMAP` header that was found was width `0x143`, height `0x158`, 8 bpp, stride `0x258`, `bmBits` null. The splash is a DIB. Which property on the page object holds its id is not known.

Lesson picture counts from the early carve: 71 BM and 5 JPEG in the lesson, 76 BM and 12 JPEG in `nida81`. BAS segment `1150` builds the book palette. An 8 bpp BMP needs that palette. Which palette a given carved BMP must use is open.

14. OpenScript bytes

14.1 What is current

Known. Handlers are compressed bytecode. The keyword pool is `TB80CMP` NE resource `0x77` (type 100, id 119). Resource `0x78` (id 120) is the other emit routine. `COMPILERPROPINFOFROMINDEX` locks resource `0x70`. Property names are a different table: BAS resource type `0x2D1`, id `0x2D0` (`bounds` `0x4004`, `idNumber` `0x4015`, `script` `0x4023`, `text` `0x402E`, `vertices` `0x4034`).

An early string dump of resource `0x77` counted 992 strings in 15776 bytes, and of `0x78` counted 980 in 15616 bytes. The decoder that actually produced the 76 Logic Families handlers counts 994 keywords. Use 994. The offset table starts at byte 2 plus count times 4. An earlier loader used count times 2 and was wrong even before the back-reference bug.

`COMPILERCOMPILETEXT` is CMP ordinal 1. `COMPILEREXPANDTEXT` is ordinal 3. `CFUNEXECUTE` is 8. `CFUNEVALUATE` is 9.

14.2 Encoding

From `FUN_1040_0970`. This is the encoding the handlers decode with.

Range	Meaning
`0x00`–`0x7F`	Literal character.
`0x80`–`0xBF`	Keyword. Index is `byte & 0xF`, or `((byte & 0xF) << 8) \`
`0xC0`–`0xFF`	Back-reference. Copy from the compressed **input** stream, not from the output.
`0xE0`–`0xEF`	CRLF plus spaces.
`0xF0`–`0xFE`	Spaces.
`0xFF`	Repeat.

The `0xC0`–`0xFF` range is also where the whitespace classes sit. The expander classifies the byte before it treats it as a generic back-reference. Do not implement "every byte above `0xBF` is a back-reference" and also implement the whitespace rows as a second pass over the same byte without the classification `FUN_1040_0970` uses.

Back-reference, as fixed on 2026-10-10:

```
if bit 4 is clear:
    length = (byte & 0xF) + 3
    src = (pos + 1) + (-4 - next_byte)
else:
    length = (byte & 0xF) + 4
    src = (pos + 1) + (-1 - word_offset)
copy `length` bytes from input[src]
```

A windowed decoder that only looks inside the slice under test will mark a back-reference that lands before the slice. On `tb80r.sbk`, reading those from the whole book dropped the `?` marks from 14256 to 66. The 66 that remain are question marks in the script.

The NIDA navigation decoder stops at the first standalone `end`. Trailing garbage is dropped. That is a corpus tool, not a claim that the expander itself stops there.

14.3 The superseded mismatch

`tbworkingpapers/tbwp0001/openscript_notes.md` says, in its first conclusion, that the decoder emits "italic", "third", "fourth", and "systime" where the script should say something else, and that the bytecode might not be OpenScript. That conclusion is superseded by the same file's later sections, and by this chapter.

Two bugs produced it. The keyword offset table was read as count times 2. Back-references copied from the output stream. After both fixes the same bytes are:

```
system lastbook
  lastbook = name of this book
  system nextpage
  nextpage = name of this page
  go to page "P20001" of book "nida81.tbk"
```

```
end
```

Other handlers that are real source: `send next`, `send previous`, `go to page "P400"`, `go to page "P310" of book "nida81.tbk"`.

Do not reopen "the keyword table is the wrong resource" from the early half of that note. Keywords live in `TB80CMP`. `TB80BAS` has no `handle` or `buttonClick` strings. `TB80RTM` has a small table at file `0xAACAD` (`handle`, `get`, `set`, `notifybefore`, `notifyafter`, `true`, `false`, `to`) and not the compiler keyword pool.

14.4 Corpus

Book	Handlers	How they are stored
`9e954d01.tbk`	76, on 39 pages	Plain `to handle` prefix, compressed body. An intermediate extract that required a plain `END` found 22. 76 is the count after the input-stream fix.
`nida81.tbk`	100	All `buttonClick`. Same encoding.
`tb80r.sbk`	33	The `to handle` / `to get` prefix is compressed too. A scan that keeps windows whose output starts with those words is how they were found. More `0xFF` and `0xE0` than the lesson books.
`nc81base.sbk`, `nc81recs.sbk`, `nc81test.sbk`, `nc81trnr.sbk`	0 decoded	High-byte runs do not decode as this bytecode. Early pattern counts (61, 17, 94, 62) were not handlers.
`tb80anm.sbk`, `tb80hyp.sbk`	0 decoded	Same. Strings and menus, not this handler format.

Lesson books store the handler name as plain text and compress the body. `tb80r.sbk` compresses the name as well. Encoding rules match. "No script at all in any system book" is too strong for `tb80r.sbk` and still true, as decoded text, for the four `nc81*` books.

14.5 What the courses actually do with it

The common subset is `go to`, `send`, assignment, `put into`, `set to`, `get`, increment and decrement, nested `if` / `else` / `end if`, `step from` / `to`, and `name of this book` / `this page`.

The clean-room engine also accepts `send NAME to TARGET`, a `get` that is a call, `forward to TARGET`, `break NAME`, and `PROP of TARGET`. Predicates such as `is not in` are recorded rather than evaluated. `local` and `system` declarations are ignored. Property writes are not applied to a live object. The expression evaluator copies `FUN_1008_2d31`: `+ - * /`, `abs`, `sqrt`, `log`, `exp`, `sin`, `cos`, `atan`, `asin`, `acos`, and otherwise leaves the original text. That function is not the record reader.

The page-by-page jumps are in chapter 13. Navigation is not "only a quoted page id." P20000 goes to P20001 because the handler says so.

14.6 Not done

A short script compiled in Instructor 8, saved, and expanded with ordinals 1 and 3 would lock the 994-name table against a known source. That test is written down and not run. Nested property expansion, opcodes past the common subset, and `forward` into a system-book handler are unimplemented or only sketched. The CMP expand host that died in `TB80W16` (`001c:00000015`, segment not present) is parked and is not the NIDA open path.

15. System books and the NIDA shell

`.sbk` and `.tbk` are the same container. Most of the Asymetrix `.sbk` files are resource containers: an object-type name table and menu resources. A few carry compressed handlers. The NIDA lesson still opens if the system books are removed. What breaks is the shell: blue screen, resize, and the menu. Renaming `nida81.tbk` removes the splash and reports that the shell is not a book. The lesson can still be entered.

15.1 Type names

The name table, in the order recovered from the system books, is: ComboBox, Stage, Picture, Hotword, RecordField, PaintObject, Curve, AngledLine, Pie, Arc, IrregularPolygon, Polygon, Line, RoundedRectangle, Ellipse, Group, Field, Button, Rectangle, then Page, Background, Book.

In `nc81base.sbk` that table sits around file `0x15E4`–`0x1ADC`. Curve's type id is `0x05C7` at file `0x1810`. The schema spelling `IrregularPloygon` is the 11.5 XSD, not this table.

These names are not yet tied to the kind byte inside a `14 00 00` chunk. `TB80UTL` holds the strings. It has not been shown to be the function that classifies a chunk.

15.2 NIDA shell

`nida81.tbk` is a real book. 43 objects. Pages `P20000` through `P22130`. It does not name lesson ids. It has 100 `buttonClick` handlers in the same bytecode as a lesson. P310 is Minimum Requirements (586, 133 MHz, 12 MB RAM, 1 GB, mouse, Windows 98). P20001 is the small menu. P22130 is Test Console Setup.

`9e954d01.tbk` at file `0xB1A96` names `tb80r.sbk`, `nc81trnr.sbk`, `nc81base.sbk`, `nc81recs.sbk`, and `nc81test.sbk`, and it calls `initializebase`, `initializerecs`, `initializetest`, `openSession`, and `gotoPageList`. It also jumps to `nida81.tbk` and to `93954302.tbk`.

Thought. Those initialize names have not been watched as a call log, and the order in `toolbook.lst` has not been watched live. `version.ini` in the set is `03.001.0006`, dated 04-18-05. `CAILT.DAT` maps course ids to Navy CDs. `9E954` maps to DIE. `STDAPPS.LST` is a short encoded list, not a book.

15.3 Asymetrix `.sbk` tiers

Sizes are the files that were triaged. JBO is at `0xC00` unless noted.

Tier	File	Size	What it is for
Core	`tb80r.sbk`	954 KB	Reader book. 66 objects. 33 compressed handlers (chapter 14). Help, bookmarking, logging, path manager, popup page manager. Not the engine.
Core	`tb80.sbk`	3.1 MB	Large core system book. Type table present. Not fully extracted.
Core	`tb80a.sbk`	2.4 MB	Author-side companion. Not on playback.
Core	`tb80i.sbk`	619 KB	Instructor system book.
Core	`INST80.SBK`	70 KB	Installer book. Referenced by Inst80. Names `tb80r`.
Useful	`tb80hyp.sbk`	141 KB	Hyperlink and help. No decoded `to handle` text.
Useful	`tb80anm.sbk`	64 KB	Path animation. No decoded `to handle` text.
Useful	`tb80aicc.sbk`	86 KB	AICC strings. Not shown to be required for a local page.
Useful	`tb80actr.sbk`	208 KB	ActiveX runtime.
Parked	`tb80acta.sbk`	2.1 MB	ActiveX author.
Parked	`tb80axf.sbk`	618 KB	ActiveX related. JBO at `0xE00`.
Not on the local path	`tb80web.sbk`	1.1 MB	Web.
Not on the local path	`tb80ftp.sbk`	243 KB	FTP. JBO at `0xE00`.
Not on the local path	`tb80x.sbk`	256 KB	Unclassified specialty.

Tier	File	Size	What it is for
Not on the local path	`tb80lmx.sbk`	62 KB	Small. JBO at `0xE00`.

An early triage said system `.sbk` files do not contain OpenScript. That is superseded for `tb80r.sbk` only. The other rows above still have no decoded handler text.

The four NIDA books `nc81base`, `nc81recs`, `nc81test`, and `nc81trnr` are password-protected resource containers. Runtime opens them without the password. The author path is the one that asks. The password is not required to read this decompile. Chapter 16 is the cipher, not the password.

15.4 Trainer and session DLLs

`nc81int.dll` and `nc81rec.dll` are 32-bit PE. They are not Asymetrix.

`nc81int.dll` exports 47 trainer calls. `IntCheckComm130E` opens a port at 9600 8O1, sends a probe, and returns `0x3D` on match or `0x3C` on failure. Families: 130E, TD2000, 5050, 205E.

`nc81rec.dll` writes `0x46`-byte records to `SESSION3.REC`. Exports include `RecStartSession`, `RecCloseSession`, `RecBookmarkSession`, `RecWriteResponse`, `RecReadVariable`. `initializeRecs` on the shell clears responses when a session loads. That last sentence is inferred from the shell script, not from a live trace of the DLL.

15.5 `linkDLL` versus a system book

A system book adds script above the book. A `linkDLL` adds commands, and only when a handler links it. The set that shows up that way is `TBDLG`, `TBWIN`, `TBDOS`, `TBREG`, `TBDB3`, `TBPDX`, and `TBCBT`. `TBWIN` is `INIT_TBKWIN` and `TBM_EVALUATE`. Neither `TBWIN` nor `TB80NET` walks page records. `TBCBT` is the CBT scoring extension and the password cipher. It is not the record reader.

16. The password cipher

The NIDA system books are password-protected. A lesson runtime opens them without presenting the password. Instructor and the author path are what ask. This chapter records where the cipher is and how the addition works. It does not record a working book password. The short strings tried against a stand-in key are not that password.

16.1 Where it lives

`TBCBT.DLL`, NE segment 7. Ghidra's name for that segment is `Code7`, base `1030`. The usual map is `0x1000 + (N - 1) * 8`, so segment 7 is `1030`. NE segment numbers stay fixed. Live selectors change. Segment 7 was not mapped in the SoftICE session that listed TBCBT code segments `01`, `0E`, `02`, `0B`, `08`, `0D`, `04`, and `03`. Those loaded segments are stubs and KERNEL/USER imports, not the cipher.

Exports on this path: `MAKEKEYEDPASSWORD`, `MAKEKEYEDPASSWORDEX`, `KEYEDPASSWORDENCRYPT`, `KEYEDPASSWORDENCRYPTEX`, `ENCRYPT`, `ENCRYPTEX`, `DECRYPT`, `DECRYPTEX`, `DECRYPTFILE`, `CREATEENCRYPTEDFILE`, `READENCRYPTEDFILE`, `WRITEENCRYPTEDFILE`.

The book path is `MAKEKEYEDPASSWORDEX` then `KEYEDPASSWORDENCRYPTEX`.

`TB80RCA.DLL` is Asymetrix UI (ASYMBITMAP, STATIC, TABS, spell, palette, BMR). It is not on the book-open path and not on the password path. `TB80ATH.EXE` is the 16-bit author and runs under NTVDM. The 32-bit pieces on a lesson launch are the launcher and the thunk (`TB80RUN`, `TB80W32`, `TB80MAIN`), not the cipher.

16.2 Three addition schemes

From `MAKEKEYEDPASSWORDEX`, `KEYEDPASSWORDENCRYPTEX`, and `FUN_1030_1046`. Inverse of each is `plain = (cipher - key) & 0xFF`.

Scheme	Add	Key byte
`(0xA01, 0x300)`	`toupper` of the key byte	From a 10-byte table. Index is `local_16 % 10`, then `+ 0x32`.
`(1, 0x400)`	`toupper` of the key byte, wrap at `0xFF`	Same table.
`(1, 0x700)`	The raw key byte, wrap at `0xFF`	Same table.

The live key is the 10 bytes that table selects. It is not the ASCII the user types. A test that encrypts `"nida"` or `"adin"` with a stand-in key `A–J` only checks the adder. It does not recover a book.

After a failed password dialog, search memory for the typed ASCII and for the constants `0xA01` and `0x300`. That finds the call site even while segment 7 is unloaded. Clicking OK and accepting "Wrong password" is the intended setup for that search. The dialog procedure itself is an ordinary Windows dialog.

16.3 What a with-password file difference is

A small book saved with a password and the same book saved without one differ from file offset `0xCEC` onward. The tails do not reconverge. The extra length is 1526 bytes of shift, not 1526 bytes of cipher text sitting at the end.

The local blob at the first difference begins:

```
c9 18 03 00 00 00 00 00 14 00 00 f0 00 00 77 2e
18 00 77 2e 6e 00 00 00 0c 45 00 00 7a 45 00 00
54 00 16 0d 00 00 0b 00 98 05 56 0e 00 00 00 00
```

The stand-in key does not decrypt it. Brute-force hits that came back short (`vgah`, `ap1`, `db9`, `iud`, `lxg`, `t57`, `vga`, `3p1`, `3ro`, `9e4`) are not the book password. They are discarded.

16.4 What is still open

The 10-byte key used on a real NIDA `.sbk` has not been captured. Segment 7 has not been stepped on a failed password. Until that capture exists, do not treat a decrypted string from the stand-in key as a finding.

17. Modules

Eight modules were counted for pattern coverage on 2026-10-11: 10,991 unique functions, 1,918 nameable by pattern (17.5 percent). Confirmed renames after the later digs are 112: BAS 20, BMP 14, RTM 9, and the rest across CMP, UTL, W16, LNL, MM, RCR, FLT, NET, JPG, and VBX. Pattern coverage: RTM 32.7 percent, UTL 41.1, BMP 26, BAS 17.4, W16 17.3, LNL 12.2, CMP 5.6. A pattern name is a candidate. The scripts in `rename/` are what was applied. Naming another thousand `StrCpy` wrappers will not find the record reader.

17.1 Launch

Piece	What it is	What it is not
`TB80RUN`	32-bit launcher. Spawns `TB80NET.EXE` with the book on the command line.	Not an in-process thunk, and not the parser.
`TB80NET.EXE`	16-bit NE. Network provider: zlib, `TBKNET3WNDPROC`, `TBK.INI`. Class `TbkNet3TB80`.	Does not parse pages.
`TB80RTM.EXE`	16-bit shell. `TBKMain`, `TBKClient`, `TBKChildWindow`, `TBKReserved_1`, `SchedulerWndTB80`. `WM_PAINT` calls `CLIENTTOPAGE`, then BAS ordinal 26.	Does not draw shapes. A lesson click arrives on `TBKClient`, not `TBKChildWindow`. `BMSG` does not see that queue.
`TB80MAIN.EXE`	23552-byte 32-bit stub. Calls `TB80RMT` `CreateRemoteCallServer` and pumps keys. Failure text: "TB80MAIN: Loading ActiveX support failed."	Not the book engine.
`tbload.exe`	Finds `ZZZTBKBOOKLOC` via `MINSTB32.dll` (`FUN_00405460`). Virtual call at `00406aa1` inside `FUN_00406a70` (vtable+4).	The callee is unresolved. Thought: first handoff out of the 32-bit search.
`TB80W32.DLL` / `TB80W16.DLL`	Thunk halves, about 246 and 406 exports. W16 is also the `DoCmd*` bus (`DoCmdDraw`, `DoCmdMouseDown`).	W16 setup (platform, OLE, 32-bit `LoadLibrary`) is not book open.
`TBFILE32.DLL`	Long filenames for the 16-bit engine.	
`TB80F32L.dll`	Flat 32-bit entry and the common-dialog hook.	
`TB80RMT`	Remote-call server used by `TB80MAIN`.	Not on the path that opens a lesson.
`PHYSSIZE`	A 1.5-era index.	Not in these 8.1 books.

The engine calls 16-bit `USER`, `GDI`, and `KERNEL`. A log that says `USER32` is missing is not a broken install. The images in hand are 16-bit NE. ToolBook 9.0 is the 32-bit product. A correspondent's claim that 8.1 is 32-bit and needs Windows 10 is rejected by the binaries (claim C-006).

17.2 Book server and neighbors

Module	Role	Not
`TB80BAS.DLL`	Objects, properties, painters, twip scale, file open. 606 exports. `CDBINIT` ordinal 1, `CDBFILEOPEN` 8, `CDBVIEWPAGE` 15, `CDBVIEWBKGND` 18, `CDBPAINTCLIENT` 26, `CDBSENDCREATE` 51 (author).	Not the window.
`TB80CMP.DLL`	Compile and expand. Resources `0x77`, `0x78`, `0x70`.	Not the page walker. 5.6 percent pattern-nameable because it is a compiler.
`TB80UTL.DLL`	Type-name strings, string manager, metrics, heaps.	Not the `14 00 00` reader.
`TB80RCR.DLL`	Linked beside the book server. Exports only partly named.	Not classified.
`TB80ENU1` / `ENU2`	English strings, 16-bit and 32-bit.	Not code that opens a book.
`TB80BMP.DLL`	DIB, metafile, FX blit. Ordinal 3 is the bitmap reader. `Lz77Expand` is `FUN_1000_0031`.	Not a picture store. The widget still does not name the bitmap.
`TB80JPG.DLL`	JPEG. Version resource still says `TB60JPG.DLL`.	
`TB80LNL.DLL`	MCI, wave, MMIO. The notify callback is Lionel.	
`TB80MM.DLL`	`tbkmmNotify`, `imageNotify`, `mciNotify`, `timerNotify`.	Does not classify objects.
`TB80OLE` / `OCX` / `ACX` / `VBX`	Clipboard, controls, automation.	Not on the Logic Families paint path.
`TB80FLT`	Filter registration. `FUN_1028_2680` calls `TBREGISTERFILTER`. The `0x14` there is a type-code limit.	Not the reader.

Module	Role	Not
TB80RCA.DLL	Author UI widgets.	Not book open, not the cipher.
TBCBT.DLL	CBT scoring. Segment 7 is the cipher (chapter 16).	Not the reader. Not loaded as segment 7 in the session that listed the other code segments.
TBDLG, TBWIN, TBDOS, TBREG, TBDB3, TBPDX	linkDLL commands.	Not page walkers.
DIBUTIL.DLL	DIB helper beside BMP.	Not the object format.

Author binaries TB80EDT.DLL and TB80.EXE are not in this set. The runtime half of the author contract is visible: BAS exports CDBEDITBOOK, CDBVIEWBOOK, CDBISVIEWERINGENERICEDITMODE, CDBSENDEDITLEAVETOALL*, and TBXAUTHORMODE. RTM calls GETAUTHORLOCALIZEDDLL. TB80ATH.EXE is the 16-bit author, seen under NTVDM. That is bookmarked, not solved.

17.3 Closed as the reader

Candidate	What it actually is
CDBSENDCREATE (ordinal 51, 12c8:00BC)	Author create and rewrite.
FUN_12c8_0ef8	Copies four words on that author path. The source is not the disk record.
FUN_1028_2680	Filter registrar.
FUN_1008_2d31 (switch 1008:2DC7)	OpenScript math.
TB80RTM	Paint handoff only.
CDBCHUNKFIND / LAST / END / CDBSKIPSEP	Text separators.
VALUENEWLONG and the 4267 constructors	Value constructors.
FUN_10b0_00a6	Type dispatcher. Code 0x14 is a type code.
12e8:0000	OLE object clone.
BAS segment 182	39 far calls with a zero segment and no reloc. Not book open.
BAS segments 1018, 1278, 1478, 14b8, 1508, 10d0, 1030, 14a8, 1160, 1070	Sampled. No stream walker. 1160 is debugging (CDBGETNEURONURL, CDBATTACHBREAKADDRESS). 1018 creates pens, brushes, and bitmaps.
A static search for the bytes 14 00 00	Thirteen undisassembled runs contain the word 14 00 as immediates, MOV CX, 0x14, or jump displacements. Not a far-pointer compare that builds an object.

17.4 Ghidra

Two NE loader bugs were fixed for this work: the entry-table segment byte was treated as signed, and ne_cbenttab was ignored. TB80BAS is the image with entry segment bytes at or above 128. toolbookunpacker clears the Optloader self-load flag and does not patch the entry table. The Ghidra patch does. An honest BAS image loads in the patched Ghidra 12.1.4 build.

Force-disassemble: walk undefined C8 (ENTER) bytes first, then the prologue pass. The prologue pass accepts a real RET / RETF before treating the following byte as code, and it only creates functions at entry-like symbols. They do not name anything.

SENDMESSAGE (321 hits in one RTM listing) and LSTRLEN (282) have mangled arguments. That is a decompiler problem. It wants a fresh decompile, not another reading of the same HTML.

maps/bas-ne-to-ghidra.md and maps/rtm-ne-to-ghidra.md are the segment crosswalks. maps/wdw-to-ghidra.md is one boot's selectors. None of the three is a narrative.

18. Where every loose sheet went

The sheets stay on disk under `tbworkingpapers/` so the trail can be checked. They are not the current statement. `who-is-who` is not in this index and is not in this repository. History folders and miscellaneous dumps are not re-copied. Binary password deltas stay in the working papers.

A row marked superseded means the sheet still contains a sentence this record closed. Read the sheet for the function body if you need it. Read this record for the claim.

Loose sheet	Fate	Where the claim lives
`tbwp0001/00-architecture.md`	kept	Chapters 3 and 17.
`tbwp0001/01-dll-catalog.md`	kept	Chapter 17.
`tbwp0001/02-open-questions.md`	superseded	Chapter 10. The reader is still unnamed; the create-path wording in that note is superseded.
`tbwp0001/ToolBook_Architecture.md`	superseded	Chapters 3–5. The book-open flowchart is superseded by chapter 4.
`tbwp0001/ToolBook_Chapter.md`	superseded	Chapters 3–5 for layout. Book-open flowchart that calls CDBSENDCREATE is superseded by chapter 4.
`tbwp0001/naming_summary.md`	kept	Chapter 8 and chapter 17. Counts 112 confirmed.
`tbwp0001/openscript_notes.md`	superseded	Chapter 14. The early italic/systime conclusion is superseded by the back-reference fix in the same file.
`tbwp0001/tbworkingpapers-pointer.md`	index	Chapter 0. Pointer only.
`tbwp0002/README.md`	kept	Chapter 7. Maps are NE-segment:offset and are not loaded as modules.
`tbwp0002/bas-create-path.md`	superseded	Chapter 4. CDBSENDCREATE is not the runtime reader.
`tbwp0003/p1040-breakdown.md`	kept	Chapter 13.1. TBX expansion of P1040.
`tbwp0003/p1040-tbk-raw.md`	kept	Chapter 13.1. Bounds, fills, blob offsets.
`tbwp0003/tb80r_compression.md`	kept	Chapters 14 and 15. 33 handlers, prefix also compressed.
`tbwp0003/tbx-vs-tbk-analysis.md`	kept	Chapters 2 and 13. Bounds in 8.1, Vertices in the export.
`tbwp0004/README.md`	kept	Chapters 2 and 5. Specimen graphics. Renders are not re-shipped here.
`tbwp0005/README.md`	kept	Chapters 6 and 9. Engine subset.
`tbwp0006/foollege-hierarchy.md`	kept	Chapter 9. Foollege is not a running server.
`tbwp0006/inventory.md`	kept	Chapter 13.7. Tag counts. Folder names are not runtime page names.
`tbwp0007/bounds-extractor.md`	superseded	Chapters 5 and 13. The 14 00 00 01 becomes 11 hypothesis in the header era is not true (WV-010).
`tbwp0007/content-rectangles.md`	kept	Chapters 5 and 13.1. Bounds-only, marker 01 00 00 04.
`tbwp0007/core-sbk-confirmation.md`	kept	Chapter 15.
`tbwp0007/denny-exchange-summary.md`	kept	Chapter 2 for the Save To XML steps. The 32-bit claim in that exchange is rejected (C-006). Personal names are not copied.
`tbwp0007/ghidra-research-list.md`	superseded	Chapters 10 and 17. Create-path items in that list are superseded.
`tbwp0007/minor-decompile-notes.md`	kept	Chapters 8 and 17.
`tbwp0007/new-dumps-analysis.md`	kept	Chapters 7 and 12. A dump is real only when it contains known book strings.
`tbwp0007/page880-comparison.md`	kept	Chapter 13. Page880 is P1040, IdNumber 880.
`tbwp0007/processing-log.md`	method	Chapter 18. A log, not a separate finding.
`tbwp0007/sbk-triage.md`	superseded	Chapter 15. The 'no OpenScript in any sbk' line is superseded for tb80r.sbk only.
`tbwp0007/schema-usefulness.md`	kept	Chapter 2. Bounds required, Vertices optional, IrregularPloygon typo.
`tbwp0007/tbk-tbx-comparison.md`	kept	Chapters 2 and 13.
`tbwp0007/tbx-schema-notes.md`	kept	Chapter 2.
`tbwp0007/template-triage.md`	kept	Chapter 15. Templates are not the NIDA playback path.
`tbwp0007/waiting.md`	kept	Chapter 10. Waiting list folded into the ranked gaps.
`tbwp0008/cdbsendcreate-sites.md`	superseded	Chapter 4. Callers are author-side.
`tbwp0009/bas-html-extracts.md`	superseded	Chapter 4. Extracts that treat ordinal 51 as runtime are superseded.
`tbwp0009/html-coverage.md`	kept	Chapter 17. HTML ?? is not absence.
`tbwp0010/group-vs-offset-table.md`	kept	Chapter 13.4. The word table is an index, not a Group.
`tbwp0011/bas-xref-walk.md`	superseded	Chapter 4. Cross-references that start at ordinal 51 are the author path.
`tbwp0011/cdbsendcreate-and-nida81.md`	superseded	Chapters 4 and 15. Author-only correction wins.

Loose sheet	Fate	Where the claim lives
tbwp0012/new-html-listings.md	kept	Chapter 17. TBCBT, TBDB3, TBDLG, tbload, TBWIN surveyed.
tbwp0012/scaletwips-and-dispatcher.md	kept	Chapters 4 and 5. FUN_1008_01dd and the type dispatcher.
tbwp0013/14-00-00-pattern-search.md	superseded	Chapter 4. No literal compare found. Author-path sentences superseded.
tbwp0013/new-html-survey.md	superseded	Chapter 17. Survey. Author-path readings superseded.
tbwp0013/paintpolygon-pointer-search.md	kept	Chapter 5. No static far pointer to 1408:0376.
tbwp0013/switch-1008-2dc7.md	kept	Chapters 6 and 14. Math, not the reader.
tbwp0013/tbload-zzztbk-walk.md	kept	Chapters 2 and 17. Marker search and the unresolved vtable+4.
tbwp0014/14-00-00-in-undisassembled.md	kept	Chapter 4. The hits are immediates and displacements.
tbwp0014/approach-correction-undisassembled.md	kept	Chapter 4. Do not close a function for lacking the literal.
tbwp0014/author-side-traces.md	kept	Chapter 4. Author exports are real and not book open.
tbwp0014/bas-seg182-leftover.md	kept	Chapter 4. Not book open.
tbwp0014/code94-ole-clone.md	kept	Chapter 4. 12e8:0000 is an OLE clone.
tbwp0014/fun-1028-2680-is-filter-registrar.md	kept	Chapter 4. Closed.
tbwp0014/high-priority-status.md	superseded	Chapter 10. The disk-to-11 expansion row is not true.
tbwp0014/instructor8-compile-test.md	kept	Chapter 14.6. Written, not run.
tbwp0014/low-priority-status.md	kept	Chapter 10.
tbwp0014/medium-priority-status.md	kept	Chapter 10.
tbwp0014/new-html-survey.md	superseded	Chapter 17.
tbwp0014/openscript-token-map.md	kept	Chapter 14. What compile-plus-expand would lock.
tbwp0014/paintpolygon-table-static-limit.md	kept	Chapter 5. Table is runtime-built.
tbwp0014/supporting-modules-content.md	kept	Chapter 17.
tbwp0014/supporting-modules-useful-for.md	kept	Chapter 17.
tbwp0014/undisassembled-regions.md	kept	Chapter 17. Forced disassembly does not name them.
tbwp0014/undisassembled-triage.md	superseded	Chapter 4. Author-path readings superseded.
tbwp0015/README.md	method	Chapter 17. C8 pass, then prologue pass.
tbwp0015/win16-merged/README.md	method	Chapter 17. Duplicate of the win16 note.
tbwp0015/win16-merged/scripts/ForceDisassembleC8.md	method	Chapter 17. Duplicate.
tbwp0015/win16-merged/scripts/ForceDisassemblePrologues.md	method	Chapter 17. Duplicate.
tbwp0015/win16/README.md	method	Chapter 17. Win16 support notes for the unpacker.
tbwp0015/win16/scripts/ForceDisassembleC8.md	method	Chapter 17. Method. Run before the prologue pass.
tbwp0015/win16/scripts/ForceDisassemblePrologues.md	method	Chapter 17. Method.
tbwp0016/openscript-expression-path.md	kept	Chapters 6 and 14. FUN_1008_2d31.
tbwp0016/page-background-index.md	kept	Chapters 4.5 and 10. Background id is not a word in the page record that has been decoded.
tbwp0016/property-block-expansion.md	superseded	Chapter 4. FUN_12c8_0ef8 is author-side. The 01-to-11 mutation is not true.
tbwp0017/bas-segment-pass2.md	kept	Chapter 4. Segments sampled, no stream walker.
tbwp0017/bas-segment-triage.md	kept	Chapter 4.
tbwp0017/side-quests-expanded.md	kept	Chapter 10.5.
tbwp0017/side-quests.md	kept	Chapter 10.5. Real, and not the reader.
tbwp0018/bas-decompile-4.md	superseded	Chapter 4. Same correction.
tbwp0018/bas-html-4-analysis.md	superseded	Chapter 4. Header correction stands: ordinal 51 is author-only.
tbwp0018/bas-html-4.md	superseded	Chapter 4. Same correction.

Loose sheet	Fate	Where the claim lives
tbwp0018/current-goals.md	kept	Chapter 10.6. Order of work.
tbwp0018/decompile-polish-todo.md	method	Chapter 17. Polish is not the reader.
tbwp0018/dibutil-analysis.md	kept	Chapter 17. DIB helper, not the object format.
tbwp0018/foolbook-research-notes.md	kept	Chapter 9.
tbwp0018/record-reader-candidates.md	kept	Chapter 4.4. Candidates closed.
tbwp0018/rtm-3-triage.md	kept	Chapter 17.
tbwp0018/rtm-decompile-2.md	kept	Chapter 17. SENDMESSAGE / LSTRLEN arguments want a fresh decompile.
tbwp0018/rtm-deep-analysis.md	kept	Chapters 5 and 17.
tbwp0018/rtm-html-4-analysis.md	kept	Chapter 17. RTM is the paint handoff.
tbwp0018/rtm-html-4.md	method	Chapter 17. Listing note.
tbwp0018/rtm-html-undisassembled.md	kept	Chapter 17. ?? runs.
tbwp0018/rtm-nameable.md	kept	Chapter 8.
tbwp0018/rtm-paint-path.md	kept	Chapters 3 and 5. CLIENTTOPAGE then ordinal 26.
tbwp0018/rtm-triage.md	kept	Chapter 17.
tbwp0018/today-summary.md	method	Chapter 10. Day log of 2026-10-14. Later corrections win.
tbwp0018/unanalyzed-functions.md	kept	Chapter 10. Still unnamed is not the same as still a candidate.
tbwp0018/undisassembled-triage-2.md	superseded	Chapter 4. Author-path readings superseded.
tbwp0019/keepers.md	kept	Chapter 11. Enter sizes. Selectors are that boot.
tbwp0019/wdw-running-map.md	kept	Chapter 12. The spine.
tbwp0019/wdw-to-ghidra.md	kept	maps/wdw-to-ghidra.md. Lookup, not typeset.
tbwp0023/rtm-ne-to-ghidra.md	kept	maps/rtm-ne-to-ghidra.md. Already in the repo. Lookup.
tbwp0024/keeper-list-decompile.md	kept	Chapter 11. Line numbers in TB80BAS.DLL.c.
tbwp0024/keeper-list.md	kept	Chapter 11. Live addresses. CDBSENDCREATE removed.
tbwp0024/ne-to-ghidra.md	kept	maps/bas-ne-to-ghidra.md. Lookup, not typeset.
tbwp0025/cdbsendcreate-is-author.md	kept	Chapter 4. The correction.
tbwp0026/bas-trace.md	kept	Chapter 12. 173 BAS shots.
tbwp0026/trace-review.md	kept	Chapter 12. Three traces, no reader.
tbwp0026/utl-trace.md	kept	Chapter 12. 53 UTL shots. No record walk.
tbwp0026/w16-trace.md	kept	Chapter 12. 11 W16 shots. No book open.
tbwp0026/wdw-break-syntax.md	kept	Chapters 7 and 12. @ not !, resume on is not a creation flag.
tbwp0026/wdw-trace.md	kept	Chapter 12.
tbwp0027/withpass-diff.bin	kept	Chapter 16. Binary stays in the working papers. Not committed.
tbwp0028/00-scaffold.md	index	Chapter 0. The scaffold this volume implements.
tbwp0028/chapters/00-how-to-read.md	kept	Chapter 0. Rewritten so this directory is the statement.
tbwp0028/chapters/01-corpus.md	kept	Chapter 1.
tbwp0028/chapters/02-formats.md	kept	Chapter 2.
tbwp0028/chapters/03-runtime.md	kept	Chapter 3.
tbwp0028/chapters/04-objects.md	kept	Chapter 4.
tbwp0028/chapters/05-graphics.md	kept	Chapter 5.
tbwp0028/chapters/06-openscript.md	kept	Chapter 6.
tbwp0028/chapters/07-debugging.md	kept	Chapter 7.
tbwp0028/chapters/08-naming.md	kept	Chapter 8.
tbwp0028/chapters/09-modern.md	kept	Chapter 9.
tbwp0028/chapters/10-open.md	kept	Chapter 10.

Dated folders already in this repository

Folder	Fate
20261012/	Earlier consolidation. Create-path wording is superseded by chapter 4. Painters and export notes still match chapters 5 and 2.

Folder	Fate
`20261014/`	Condensed working papers. `02-still-open.md` still calls the reader the caller of `CDBSENDCREATE`. That sentence is superseded. The rest of its open list matches chapter 10.
`20261015/`	`00-how-toolbook-operates.md` section 2 and `01-theories-of-operation.md` section 1 still teach the author chain as runtime. Superseded by chapter 4.
`20261016/00-consolidated.md`	The short correction. This volume is the expansion.
`rename/`	The scripts chapter 8 refers to.
`maps/`	Segment crosswalks. Not typeset in the PDF.
`nehost/`	Unicorn experiments. Not part of this record and not committed from the working tree unless asked.